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Original Articles

Targeting AXL can effectively overcome c-Met-induced therapeutic resistance in renal cancer and promote tumor cell death through increased oxidative stress☆

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ABSTRACT

The mechanisms underlying therapeutic resistance to c-Met/receptor tyrosine kinase (RTK) inhibitors in renal cancer remain unexplored. In renal cell carcinoma (RCC) cells, both AXL and c-Met are highly upregulated. Notably, we found that prolonged treatment with the c-Met/RTK inhibitor, cabozantinib (Cabo), a standard treatment for advanced-stage RCC, markedly increased total c-Met levels and promoted renal cancer cell proliferation. This effect was confirmed not only *in vitro* but also in murine models and renal tumor tissues from Cabo-treated patients. At lower concentrations (1 nM and 10 nM), Cabo treatment failed to inhibit HGF (c-Met ligand)-induced c-Met phosphorylation. Instead, it further enhanced receptor phosphorylation and downstream signaling events for tumor growth. Additionally, Cabo treatment induced AXL-c-Met association and disrupted the physiological degradation of c-Met. However, inhibition or knockout of AXL could significantly overcome therapeutic resistance to c-Met inhibitor(s). It triggered apoptotic cell death through increased oxidative stress and inhibition of the redox-sensitive transcription factor, Nrf2 and its effector molecule, heme oxygenase-1 (HO-1). We also generated Cabo-resistant RCC cells and observed a marked upregulation of both c-Met and AXL in these cells. Epigenomic profiling revealed significant differences between Cabo-resistant and Cabo-sensitive RCC cells. Importantly, inhibition of AXL either using a potent inhibitor, TP-0903, or through genetic silencing resensitized the resistant cells to Cabo-induced cell death. Together, our findings highlight AXL as a key driver of therapeutic resistance to c-Met inhibitors. A combination therapy targeting both c-Met and AXL in renal cancer could be a promising strategy to overcome the acquired resistance to c-Met inhibitors through increased oxidative stress.

☆ MM and LR are co-second authors with equal contributions; AS, MB, TKC and SP are senior authors.

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1. Introduction

Patients with advanced renal cell carcinoma (RCC) have limited treatment options [1]. Current therapies include anti-angiogenic agents, immune checkpoint inhibitors and tyrosine kinase inhibitors. However, despite initial responses, most patients eventually develop resistance, leading to disease progression [2–4]. Therefore, identifying novel therapeutic targets and exploring new combination treatment strategies are critical.

One such target is the c-Met receptor tyrosine kinase (RTK), which is upregulated and deregulated in renal cancer [5–7]. Upon binding of its specific ligand, hepatocyte growth factor (HGF), c-Met undergoes phosphorylation, activating tumor-promoting pathways in both clear cell and papillary RCC [8,9]. This process involves c-Met dimerization, followed by autophosphorylation at key tyrosine residues (Y1234 and Y1235), which is essential for activation. Subsequent phosphorylation at Y1349 facilitates the recruitment of adaptor molecules, such as, growth-factor-receptor-bound protein 2 (Grb2)-associated binder 1 (Gab1). Furthermore, to attenuate or regulate c-Met-signaling, Y1003 is phosphorylated, that allows the recruitment of c-CBL, an E3 ligase that mediates receptor ubiquitination [10–12]. Ubiquitination of c-Met initiates clathrin-mediated endocytosis and internalization of the c-Met dimer, which can either be routed to proteasomes or lysosomes for degradation or recycled back to the cell membrane [10,13,14].

Cabozantinib (Cabo), a c-Met/RTK inhibitor, is approved for the treatment of RCC [15]. Although it was initially developed to specifically inhibit c-Met and VEGFR2, it was later found that it can also target additional RTKs (such as AXL, RET, KIT and FLT3) at higher concentrations, albeit less efficiently. Despite initial efficacy, resistance to Cabo eventually emerges [16]; and multiple mechanisms may contribute to this resistance [17,18]. Many chemotherapeutic agents can mediate their cytotoxic functions through generation of reactive oxygen species (ROS). Our previous studies demonstrated that c-Met activation protects RCC cells from ROS-induced oxidative stress and cytotoxicity by inducing anti-oxidant factors, such as Nrf2, and the redox-sensitive cytoprotective molecule, heme oxygenase-1 (HO-1) [19]. Additionally, c-Met amplification has been implicated in therapeutic resistance in lung cancer [20], where alternative pathways, such as EGFR, is activated [21]. Importantly, c-Met can have cross-talk with other RTKs.

AXL, another RTK upregulated in both clear cell and papillary RCC [22,23], belongs to the TAM family of RTKs (TYRO3, AXL and MERTK) [24]. AXL is also considered a mesenchymal marker in few other cancer cell types [25–27]. Upon binding to its ligand, GAS6, AXL undergoes phosphorylation [28], which subsequently activates tumor-promoting pathways [29]. Importantly, AXL can have cross-talk with c-Met and it can mediate ligand (HGF)-independent activation of c-Met [30]. Notably, AXL is more potent than c-Met in driving tumor metastasis and may play a key role in therapeutic resistance [31,32]. Molecular studies on recurrent tumors have highlighted the complex role of AXL in promoting resistance to various treatments, both conventional and targeted therapies [33]. In ovarian cancer AXL has been identified as a key mediator of resistance to cisplatin. Similarly, in acute myeloid leukemia (AML), chemotherapy-resistant patient-derived cells exhibit significantly elevated AXL expression [34]. Furthermore, AXL has been implicated in resistance to cetuximab in non-small cell lung cancer (NSCLC) and head and neck squamous cell carcinoma (HNSCC) [35], highlighting its role across multiple treatment modalities.

Despite these findings, the mechanisms involved in the development of therapeutic resistance to c-Met inhibitors in RCC remain largely unexplored. In this study, we demonstrate that both AXL and c-Met are significantly upregulated in RCC cells compared to normal renal epithelial cells, forming a complex that contributes to resistance. Prolonged treatment with c-Met/RTK inhibitors, including Cabo, leads to increased total c-Met levels and promotes renal cancer cell proliferation. Moreover, lower doses of Cabo failed to inhibit HGF-induced c-Met phosphorylation; instead, they enhanced receptor activation and downstream signaling events involved in tumor growth. However, AXL inhibition or knockout effectively overcomes therapeutic resistance against c-Met inhibitors; and markedly induced apoptotic cell death through increased oxidative stress. By utilizing Cabo-resistant RCC cells, we found that both c-Met and AXL expression levels were significantly elevated. Our in-depth studies (both *in vitro* and *in vivo*) suggest that AXL plays a central role in mediating therapeutic resistance to c-Met inhibitors. Importantly, combination therapy using c-Met inhibitors along with AXL inhibitors, such as TP-0903, could be a promising strategy to overcome acquired resistance in RCC. Our study identifies a novel mechanism, by which Cabo treatment might contribute to acquired drug resistance through increased accumulation of the target receptor. These findings provide new insights into the action of c-Met inhibitors and highlight the potential of combination therapies in RCC treatment.

2. Materials and Methods

2.1. Reagents

Cabozantinib (XL-184) (Cat. No. S1119), Crizotinib (Cat. No. S1068), PF-04217903 (Cat. No. S1094), TP-0903 (Cat. No. S7846), YK-4-279 (Cat. No. S7679) and MG132 (Cat. No. S2619) were purchased from Selleckchem (Houston, TX). Some of the results from our *in vitro* studies were also confirmed using XL-184 obtained from Exelixis. siRNA reagents, dilution buffer (Cat. No. sc29527), Transfectin medium (Cat. No. sc29528), and Transfectin reagent (Cat. No. sc-29528) were purchased from Santa Cruz Biotechnology. Collagenase Type IV (Cat. No. 0709) was obtained from Stem Cell Technologies, and lymphocytes separation medium (Cat. No. 25-072-CV) was purchased from corning. The eBioscience™ Annexin V Apoptosis Detection Kit was purchased from Thermo Fisher Scientific (Cat. No. 88-8007-74). NE-PER™ Nuclear and Cytoplasmic Extraction Reagents were purchased from Thermo Fisher (Cat. No. 78833). Recombinant Human Hepatocyte Growth Factor (HEK293-derived) was obtained from PeproTech (Cat. No. 100-39H-100UG). The Click-iT™ EdU Cell Proliferation Kit for Imaging, Alexa Fluor™ 488 dye, were purchased from Thermo Fisher (Cat. No. C10340). VECTASHIELD® Antifade Mounting Medium with DAPI (#H-1200-10) was purchased from Vector Laboratories.

2.2. Cell lines

Human renal cancer cell lines (786-O, ACHN, and Caki-1) and the mouse renal cancer cell line (RENCA) were obtained from the American Type Culture Collection (ATCC, Rockville, MD). All human cell lines were reauthenticated through short tandem repeat profiling from ATCC. The 786-O, Caki-1, and RENCA cell lines were maintained in RPMI 1640 medium, while ACHN cells were maintained in Eagle's minimum essential medium. All culture media were supplemented with 10 % heat-inactivated fetal bovine serum and 1 % penicillin/streptomycin antibiotics (Invitrogen, Carlsbad, CA). Normal renal proximal tubular epithelial cells (RPTEC) were purchased from Lonza and cultured in renal epithelial cell basal medium with SingleQuots growth supplements (Lonza, Portsmouth, NH). Cabozantinib (Cabo)-resistant Caki-1 cells (Caki-1R) were generated by growing Caki-1 cells in McCoy's 5A (Modified) Medium (Gibco) supplemented with 10 % fetal bovine serum (Sigma), 100 μg/mL streptomycin (Invitrogen), and 5 μM cabozantinib, with an incremental increase over 2 months until reaching 20 μM cabozantinib. Once the cells became resistant, they were cultured in the presence of 10 μM cabozantinib.

2.3. Western blot

Western blot was performed using primary antibodies against p-c-Met Y1234/1235, p-c-Met Y1349, p-c-Met Y1003, c-Met, AXL, p-ERK1/2, ERK1/2, p-Akt, Akt, p-Stat3, Stat3, p-H2A.X, Bcl2, HO-1, Nrf2, GAPDH, Ubiquitin, Rad51, Sp1 and β-Actin (detailed information in

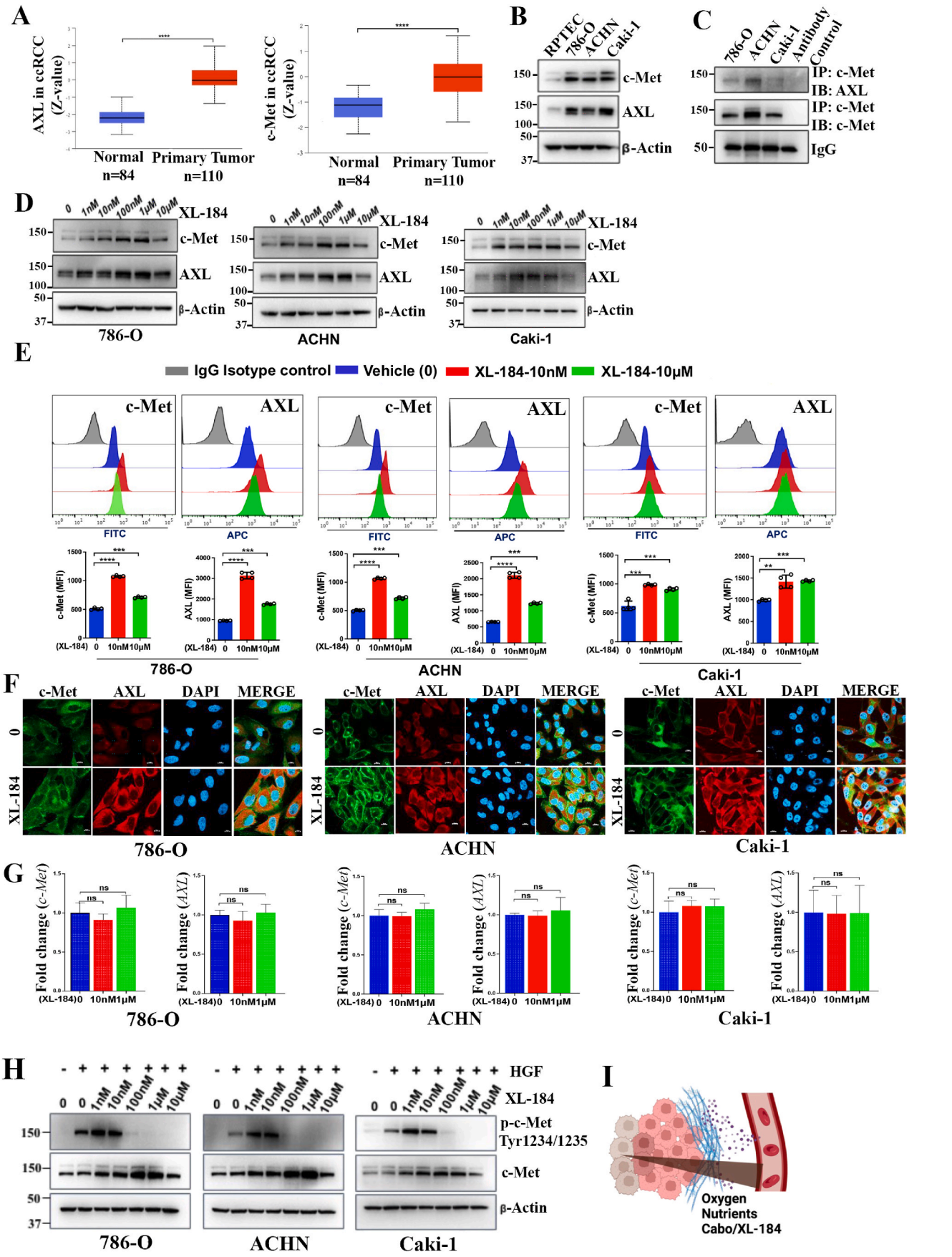

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Fig. 1. c-Met and AXL are overexpressed in RCC, and XL-184 further elevates their levels in renal cancer **cells. (A)** (A) Box plots showing mRNA expression levels of c-Met (left panel) and AXL (right panel) in normal kidney tissue and primary tumor samples from patients with ccRCC. Data were obtained from The Cancer Genome Atlas (TCGA) via the UALCAN database (http://ualcan.path.uab.edu). Statistical significance was determined using a Student's t-test; ****$p < 0.0001$. **(B)** Expression of c-Met and AXL in normal RPTEC cells versus RCC cells (786-O, ACHN, and Caki-1) was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three experiments. **(C)** Cell lysates from 786-O, ACHN, and Caki-1 cells were subjected to immunoprecipitation (IP) assay using an anti-c-Met, followed by Western blot (IB) with either anti-AXL or anti-c-Met. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments. **(D)** 786-O, ACHN, and Caki-1 renal cancer cells were treated with increasing concentrations of XL-184 (1 nM–10 μM) or vehicle (control, indicated as 0) for 24 h. Following treatments, the expression of c-Met, AXL, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments/biological replicates. **(E)** Cells were treated with either vehicle or low (10 nM) and high (10 μM) concentrations of XL-184. Following treatments, the extracellular levels of c-Met and AXL were analyzed by flow cytometry (along with isotype control), and the quantifications are presented as bar graphs (below panel). Data represent the mean ± SD of four readings, representative of three different experiments. ****$p < 0.0001$, ***$p < 0.001$, **$p < 0.01$. **(F)** 786-O, ACHN, and Caki-1 cells were treated with 10 nM XL-184 for 24 h. After treatments, cells were stained with c-Met and AXL antibodies, followed by a specific secondary antibody and DAPI. Pictures were taken with a confocal microscope. **(G)** 786-O, ACHN, and Caki-1 cells were treated with 10 nM and 1 μM of XL-184 for 24 h, and RT-PCR assay was performed. Bar graphs represent the mean fold changes in mRNA expression from three different experiments. **(H)** 786-O, ACHN, and Caki-1 cells were treated with increasing concentrations of XL-184 (1 nM–10 μM) or vehicle for 24 h (as in D). After 24 h, cells were treated with or without HGF (50 ng/mL) for 20 min. Following treatments, the expression of phospho-c-Met (Tyr1234/1235), total c-Met, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments. **(I)** Schematic representation of gradients of oxygen, nutrient, and drugs within the tumor microenvironment. The illustration highlights how the uneven distribution of these factors creates distinct zones within the tumor, leading to differential drug exposure and variable treatment outcomes.

Supplementary Table 1). Briefly, cultured cells were lysed in lysis buffer (Cell Signaling Technology). Samples were subjected to SDS-PAGE and transferred to PVDF membranes. Blots were probed with primary antibodies, followed by horseradish peroxidase-conjugated secondary antibodies. Antigen-antibody complexes were detected using an enhanced chemiluminescence reagent. Blots were scanned on a BioRad ChemiDoc MP imaging system.

2.4. Quantitative real-time PCR

Total RNA was extracted following the manufacturer's protocol using the RNeasy isolation kit (Cat. No. 74106, Qiagen). cDNA was synthesized using the High-Capacity cDNA Reverse Transcription kit (Cat. No. 436884, ThermoFisher Scientific). Quantitative real-time PCR was performed using a reaction mixture containing TaqMan 2X Universal PCR Master Mix (Cat. No. 4304437, ThermoFisher Scientific). Primers for c-Met (Cat. No. Hs01565584_m1), AXL (Cat. No. Hs01064444_m1), and an internal control, 18s (Cat. No. Hs99999901_s1) mRNA were obtained from ThermoFisher Scientific, USA, and amplified and analyzed under identical conditions. Melting point analysis was done in all cases to confirm the right amplification of the expected gene. C_t (the cycle number at which emitted fluorescence exceeded an automatically determined threshold) for gene of interest was corrected by the C_t for 18s and expressed as ΔC_t. Data were measured as fold changes of mRNA amount, which was calculated as follows: (fold changes) $= 2^X$ [where $X = \Delta C_t$ (for control group) $- \Delta C_t$ (for experimental group)].

2.5. Confocal microscopy

Cells were cultured on sterile coverslips in a 6-well plate. After treatments, the cells were fixed in 4 % paraformaldehyde and permeabilized with 0.25 % Triton X-100 in phosphate-buffered saline (PBS) for 10 min at room temperature (RT). The cells were then blocked in 3 % Bovine Serum Albumin in PBST (0.1 % Tween 20) for 1 h at room temperature with gentle shaking. Antibodies for c-Met and AXL were applied as 100 μL drops onto respective coverslips. These coverslips were then incubated overnight at 4 °C in hydration chambers. A coverslip incubated without any primary antibody served as a control. The next day, the coverslips were returned to their respective wells and washed thrice with PBST. Secondary antibodies, anti-mouse IgG and anti-rabbit IgG, were added in blocking buffer at a 1:200 dilution. The coverslips were then incubated in the hydration chamber for an hour at room temperature. The coverslips were then washed three times before being returned to their wells. Coverslips were mounted, allowed to dry, and stored at 4 °C until imaging. Imaging was performed using the Zeiss LSM 980 confocal microscope at the DFCI Molecular Imaging Core.

2.6. Generation of CRISPR/Cas9-mediated knockout cells

Renal cancer cells (786-O) were co-transduced with lentiviral particles encoding Cas9 and gRNA targeting either AXL or a control (Thermo Fisher, Waltham, MA). The stable transduced clones were selected using antibiotic-containing media and confirmed through Western blot analysis to check for the expression of AXL. Clones exhibiting verified AXL knockout (KO) were chosen for experiments.

2.7. Cell proliferation assay

Control clones and AXL-KO cells were seeded in 96-well plates at a density of 1×10^4 cells/well and incubated overnight. The cells were treated with Cabo/XL-184 (10 nM–10 μM) or vehicle for 55 h for and images were acquired hourly using a live cell imaging system (IncuCyte® S3). Cell confluency, as surrogate for cell proliferation, was analyzed by using the IncuCyte® live-cell analysis software. For the Edu cell proliferation assay, cancer cells were cultured and treated in a similar way as mentioned above. Following treatments, the cells were collected through trypsinization and were stained with Click-iT™ EdU Cell Proliferation Kit using manufacturer's protocol. Cells were analyzed by flow cytometry on a BD FACSCelesta cell analyzer.

2.8. Reactive Oxygen Species (ROS) generation assay

Briefly, 1×10^5 cells were seeded in 6-well tissue culture plates and allowed to grow in complete culture medium for 24 h. Cells were serum starved for 12 h to synchronize the cell growth. Following treatments, cells were trypsinized and stained using ROS-ID total ROS detection kit from Enzo Life Sciences (Plymouth, MA), according to the manufacturer's protocol. Subsequently, flow cytometry was performed using a BD FACSCelesta cell analyzer.

2.9. Apoptosis assay

Briefly, 1×10^5 cells were seeded in 6-well tissue culture plates and allowed to grow in complete culture medium for 24 h. The next day, the spent media was removed, and the wells were washed with phosphate-buffered saline. Treatments were performed in serum-free medium containing cabozantinib and TP-0903 at concentrations shown in the figures. Following treatments, the cells were stained with an Annexin V-allophycocyanin (APC)/propidium iodide solution according to the manufacturer's protocol (Thermo Fisher Scientific). The cells were analyzed by flow cytometry on a BD FACSCelesta cell analyzer.

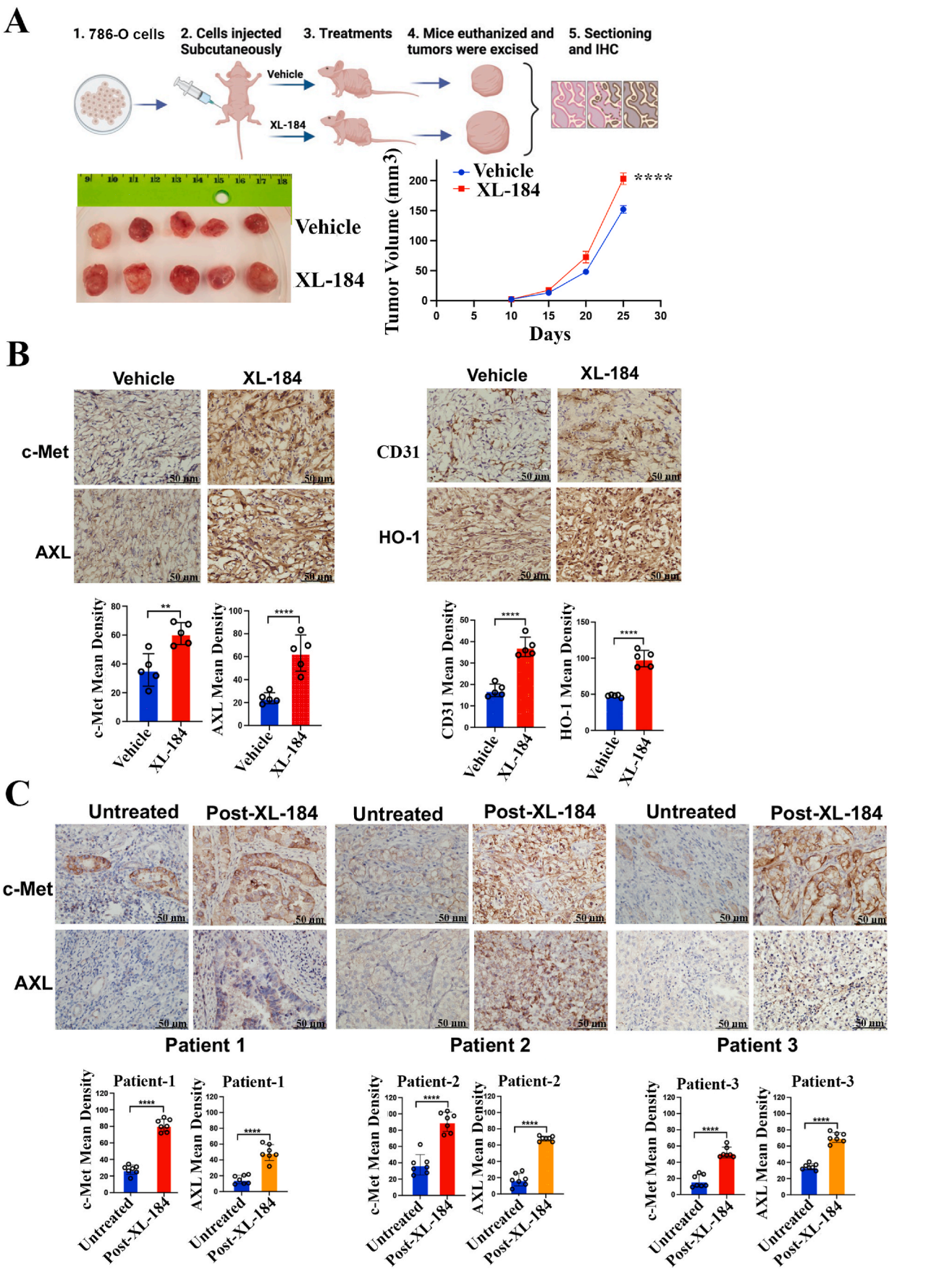

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Fig. 2. Low dose of XL-184 treatment increases renal tumor growth in a tumor xenograft **model. (A)** 786-O cells were subcutaneously injected into the flanks of athymic nude mice. Once palpable tumors were formed, mice were injected intraperitoneally (IP) with XL-184 (7.5 mg/kg) or vehicle (n = 5 per group) for ~3 weeks, and tumor volumes were measured. Representative photographs of tumors excised on the 25th day following tumor injection (left panel). The line graph shows tumor volumes recorded at the indicated time points (right panel). ****$p < 0.0001$ compared with the vehicle-treated control. **(B)** Renal tumor sections obtained from the above treatment groups were processed for immunohistochemistry. The figure illustrates representative photomicrographs (magnification × 400) showing immunohistochemical expression (brown) of c-Met, AXL, CD31, and HO-1 in harvested tumor tissues. Columns represent the average intensity values of stained proteins; bars represent ±SD of five different readings. **$p < 0.01$, and ****$p < 0.0001$. **(C)** Renal tumor tissue sections obtained from kidney cancer patients pre- (untreated) and post-XL-184 treatment, were processed for immunohistochemistry (brown) and stained for c-Met and AXL. Magnification ×400. Columns represent the average intensity values of stained proteins; bars represent ±SD of five different readings. ****$p < 0.0001$ compared with untreated as indicated. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.10. Immunoprecipitation assay

Immunoprecipitations were performed using 500 μg of cellular proteins and antibody (2 μg per 500 μg of total protein) with anti-c-Met. Immunocomplexes were captured using protein A/G-PLUS-Agarose beads (Santa Cruz Biotechnology, TX) and washed three to four times. The precipitated proteins were analyzed by Western blot using anti-AXL and anti-c-Met.

2.11. Nuclear and cytosolic fractionation

Cells were treated as mentioned for apoptosis assay. Following treatments, cells were lysed and nuclear/cytoplasmic fractions were separated using NE-PER Nuclear and Cytoplasmic Extraction Reagents Kit (Thermo Fisher Scientific, Waltham, MA) according to the manufacturer's protocol.

2.12. Xenograft model for renal tumor growth

To investigate renal tumor growth, renal cancer cells were subcutaneously injected into the dorsal flank of 8-week-old male athymic nude mice and observed for tumor formation. The mice were then divided into experimental groups (n = 5 mice per group) treated with either the vehicle or cabozantinib (Cabo/XL-184). Once palpable tumor growth was observed, mice were treated every day for three weeks via intraperitoneal injection (IP). Tumor volumes were measured using a digital caliper and calculated using the standard formula: $V = 1/2(h \times w \times w)$. The excised tumors were promptly snap-frozen in OCT (Sakura Finetek, Torrance, CA) and stored at −80 °C until further use. All animal experiments were conducted with approval from the Animal Care and Use Committee at Boston Children's Hospital (BCH) and in accordance with the National Institutes of Health (NIH) guidelines for the care and use of laboratory animals. The BCH approval number for animal studies is 00002180.

2.13. Syngeneic model for renal tumor growth

For syngeneic renal tumor growth, we utilized murine renal cancer cells (RENCA) in immunocompetent Balb/c mice. RENCA cells were injected into the dorsal flank of 8-week-old male Balb/c mice and observed for tumor formation. Once palpable tumor growth was observed, mice were then divided into four treatment groups (n = 5 mice per group) treated with vehicle, Cabo/XL-184 (7.5 mg/kg), TP-0903 (10 mg/kg) and Cabo + TP-0903 for three weeks via intraperitoneal injection (IP). Tumor volumes were measured as mentioned above for the xenograft model.

2.14. Isolation of Tumor Infiltrating Lymphocytes (TILs) and tumor cells

Briefly, excised mouse tumors from Balb/c mice were placed in 50 mL conical tube containing 10 mL fresh RT HBSS media supplemented with 0.7 mg/mL collagenase IV for 30 min at 37 °C. After digestion, contents were filtered through a 70 μm strainer into a new 50 mL conical tube followed by Ficoll-Paque Density Gradient Centrifugation method [36]. TILs and tumor cells were isolated and stained with antibodies. Tumor cells were stained with c-Met, AXL and PD-L1 and TILS were stained with CD3, CD4, CD8, PD1, NK1.1, Tim3, CD69 and FOXP3 (detailed information in Supplementary Table 1).

2.15. Tissue samples

Paraffin embedded tissue sections of human renal cell carcinoma (RCC) from patients (before and after cabozantinib treatment), were obtained TAPCD (Tissue Acquisition, Pathology and Clinical Data) Core of Dana Farber/Harvard Cancer Center. The protocol for obtaining tissue samples was approved by the hospital's review board. The IRB numbers for human tissue studies are 01–130, 07–336 and 20000.

2.16. Immunohistochemistry

For tumor xenografts, briefly, the harvested tissues were immediately snap-frozen in OCT (Sakura Finetek, Torrance, CA). After blocking in 10 % goat serum, the sections were incubated with primary antibodies against CD-31, c-Met, AXL and HO-1. The sections were then incubated with species-specific horseradish peroxidase-conjugated secondary antibody (Jackson ImmunoResearch, Baltimore, MD). All the tissue sections were washed thoroughly in between incubations. DAB was used as a substrate, and Gill's hematoxylin (Fisher Scientific, Waltham, MA) was the counterstain. Photographs were taken on a Nikon Eclipse photomicroscope. Vessel densities were quantified by grid-counting method at × 400 magnification. DAB intensity was quantified using ImageJ software (v. 1.54f), and bar graphs were plotted to show the mean intensity of DAB.

Paraffin-embedded tissue sections (human renal tumors and from Balb/c mice) were processed following standard protocols [37]. Mouse tissue sections were stained with primary antibodies against phospho-c-Met, c-Met, AXL, HO-1, CD31, CD4, CD8 and CD11c. Human tissue sections were stained with c-Met and AXL, as detailed in Supplementary Table 1.

2.17. Transfection of siRNA

Negative control siRNA (Thoroughly tested and validated non-silencing siRNA with no homology to any known mammalian gene, Qiagen, GeneGlobe ID - SI03650318), human *AXL* siRNA (Predesigned siRNA directed against Human AXL, (NM_001699, NM_021913) GeneGlobe ID - SI00131355) were purchased from Qiagen, USA. Cells were seeded in 6-well plate at 1×10^5 cells/well and incubated overnight in an antibiotic free media. Transfection was performed according to the manufacturer's protocol and the gene knockdown was confirmed by Western blot.

2.18. RNA sequencing (RNA-seq)

RNA sequencing was performed on two replicates each of the parental Caki-1 (CaboS) and CaboR cell lines using Illumina Novoseq at Novogene. The RNA reads were aligned to GRCh38 reference genome using STAR [38], quantified using RSEM [39] and evaluated for quality using RNA-seqQC2 [40]. Differential gene expression between the CaboR and CaboS cell lines was conducted using the R package

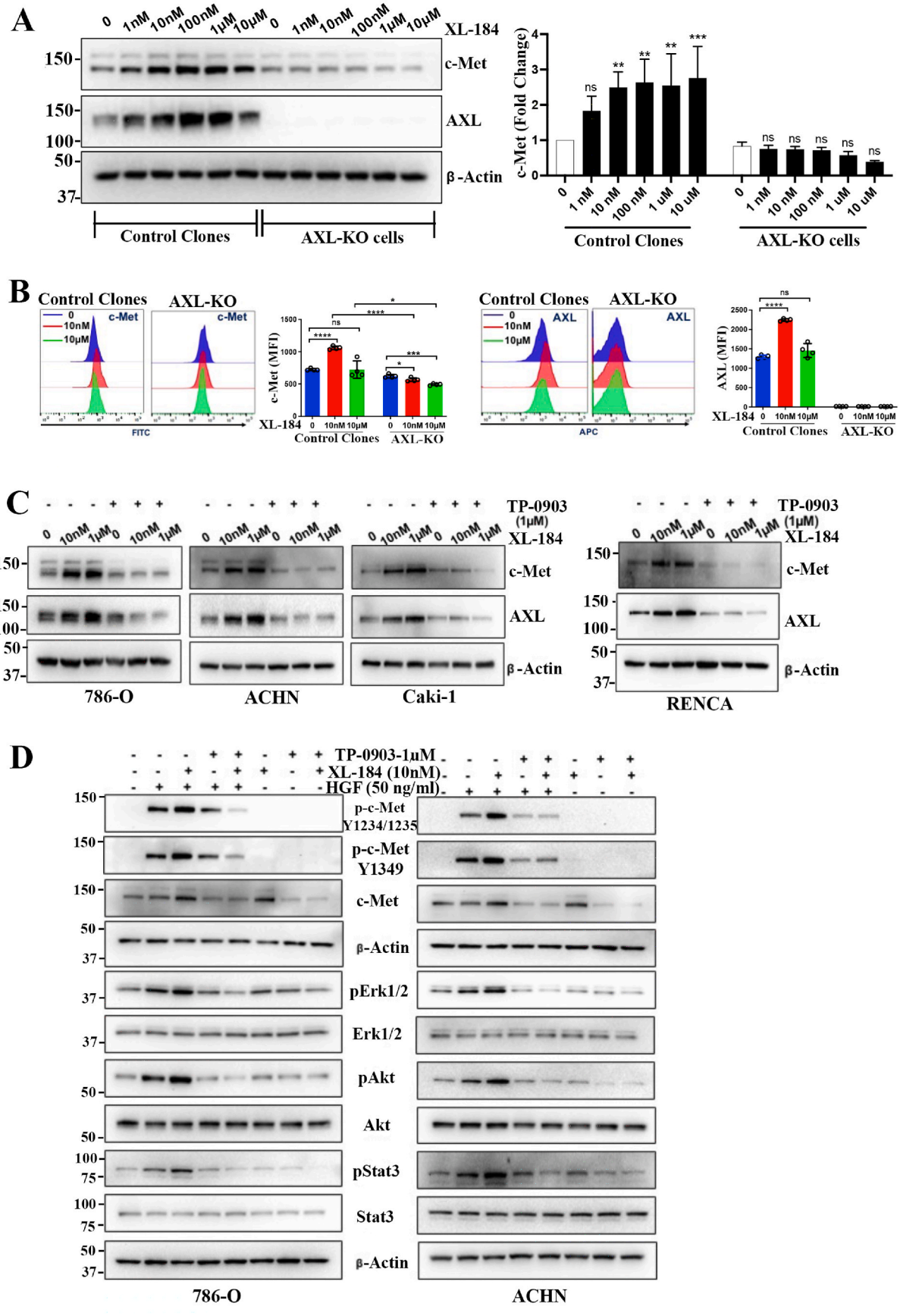

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Fig. 3. Silencing of AXL prevents XL-184-mediated c-Met overexpression; and the combination treatment with the AXL inhibitor, TP-0903, enhances the anti-tumorigenic potential of **XL-184. (A)** CRISPR/Cas9-mediated AXL-KO clonal 786-O cells (AXL-KO cells) were generated along with a control clone using lentiviral particles encoding Cas9 and AXL-specific gRNA, as described in the Materials and Methods. Control clones and AXL-KO cells were treated with different concentrations of XL-184 (1 nM–10 μM) or vehicle (Control, indicated as 0) for 24 h. Following treatments, the expression of c-Met, AXL, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments/biological replicates. The bar graphs represent the fold changes in c-Met expression by densitometry where the signals were normalized to an internal control β-actin. The columns represent the mean±SD of readings of three different samples. ***$p < 0.001$ and **$p < 0.01$ compared with controls. **(B)** Control clones and AXL-KO were treated similarly as (A) with either vehicle or low (10 nM) and high (10 μM) concentrations of XL-184. Following treatments, the surface levels of c-Met and AXL were analyzed by flow cytometry, and the quantifications are presented as bar graphs (side panels). Data represent the mean ± SD of four readings, representative of three different experiments. *$p < 0.05$, ****$p < 0.0001$. **(C)** Human (786-O, ACHN, and Caki-1) and murine (RENCA) RCC cells were treated with XL-184 (10 nM & 1 μM), TP-0903 (1 μM) alone and in combination, or vehicle (indicated as 0) for 24 h. Following treatments, the expression of c-Met, AXL, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments. **(D)** 786-O (left panel) and ACHN (right panel) cells were pre-treated with XL-184 (10 nM), TP-0903 (1 μM), or their combination for 24 h. After 24 h, cells were treated with or without HGF (50 ng/mL) for 20 min, as indicated. Following treatments, the expression of phospho-c-Met (Tyr1234/1235), phospho-c-Met (Tyr1349), c-Met, phospho-Erk1/2 (Thr202/Tyr204), Erk1/2, phospho-Akt (Ser473), Akt, phospho-Stat3 (Tyr705), Stat3, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments/biological replicates.

"DESeq2" (v.1.40.2) [41]. Genes that were not expressed in two or more samples or had an average expression of less than 10 across all samples were filtered out before the analysis. A gene set enrichment analysis (GSEA) [42] was performed using the R package "clusterProfiler" (v.4.8.2) [43] on selected pathways from the Hallmark, KEGG, Reactome, Biocarta and WikiPathways.

2.19. Chromatin immunoprecipitation (ChIP)

Chromatin immunoprecipitation and sequencing (ChIP-seq) was performed on the cell lines for the post translational histone modifications H3K27ac and H3K4me3. Both cell lines were fixed using 1 % formaldehyde (ThermoFisher) for 10 min at room temperature and then quenched with 125 mM Glycine. Chromatin was sheared to 300 to 500bp using the Covaris E220 ultrasonicator. Resulting chromatin was incubated overnight at 4^{o} C with the following antibodies coupled with protein A and protein G beads (Invitrogen): H3K27ac and H3K4me3. Five percent of the sample was not exposed to antibody and was used as a control. The beads were then washed three times each with Low-Salt Wash Buffer (0.1 % SDS, 1 % Triton X-100, 2 mM EDTA, 20 mM Tris-HCl pH 7.5, 150 mM NaCl), High-Salt Wash Buffer (0.1 % SDS, 1 % Triton X-100, 2 mM EDTA, 20 mM Tris-HCl pH 7.5, 500 mM NaCl), and LiCl Wash Buffer (10 mM Tris pH 7.5, 250 mM LiCl, 1 % NP-40, 1 % Na-Doc, 1 mM EDTA) and rinsed with TE buffer (pH 8.0) once. The samples were then de-cross-linked, treated with RNase and proteinase K, and DNA was extracted (Qiagen). DNA sequencing libraries were prepared from purified input and IP sample DNA using the ThruPLEX DNA-seq Kit (TakaraBio). Libraries were sequenced on an Illumina platform to generate 150-bp paired-end reads (Novogene).

2.20. ChIP-sequencing data analysis

Paired-end sequencing reads were trimmed and aligned to the hg19 human reference genome using Burrows–Wheeler Aligner version July 0, 1740 [39]. Non-uniquely mapping and duplicate reads were discarded. Peak calling was performed using MACS version 2.1.1.2014061641 with a q value (false discovery rate (FDR)) threshold of 0.01 [44]. Fragment locations were inferred from mapped read pairs in BAM files and converted to BED format using BEDTools (version 2.29.2) bamtobed with the -bedpe flag set [45]. Fragments were imported as GRanges objects using R package rtracklayer [46] (version 1.56.1) and resized to 1 bp at the center of the fragment location to ensure that fragments can map to only one site when assessing overlap with genomic regions of interest and overlap peaks was assessed using GenomicRanges [47] (version 1.46.1). Peaks were considered overlapping if they shared one or more base pairs. ChIP-seq heatmaps were generated with deepTools v3.3.1 [48] and show normalized read counts at the peak center ±2 kb unless otherwise noted.

2.21. Statistical analysis

Data are presented as mean ± SD. Statistical analysis was performed using GraphPad Prism 8.0.1. Student's *t*-test was used to compare two groups of data. Comparisons between multiple groups were performed using one-way ANOVA followed by Bonferroni's multiple comparison and Kruskel-Wallis test. Data differences between groups were separated into two independent factors using two-way ANOVA followed by Tukey's post hoc tests. Correlation analysis was used to analyze the correlation between two groups of data. $P < 0.05$ was considered statistically significant.

3. Results

3.1. Treatment with c-Met/RTK inhibitor increases the level of total c-Met in renal cancer cells

According to the UALCAN database [49,50], both c-Met and AXL are significantly overexpressed in primary kidney tumors, including both clear cell and papillary subtypes (Fig-1A and Supplementary Fig-1A). We confirmed this observation by showing that renal cancer cells, specifically, from clear cell and papillary types, markedly overexpress both c-Met and AXL compared to normal renal epithelial cells and form a complex as shown by immunoprecipitation assay (Figs: 1B and C). Additionally, differential expression analysis using the clinical proteomic tumor analysis consortium (CPTAC) data, accessed through UALCAN, revealed that both c-Met and AXL are overexpressed in clear cell renal cell carcinoma (ccRCC) (Supplementary Fig-1B).

Next, we investigated the effects of treatment with the c-Met/RTK inhibitor cabozantinib (Cabo/XL-184) on c-Met and AXL expression in renal cell carcinoma (RCC) cells. Interestingly, following XL-184 treatment (ranging 1 nM to 10 μM), we observed a significant increase in the levels of both total c-Met and AXL in all human RCC cell lines tested (786-O, ACHN and Caki-1), compared to controls (Fig-1D). A similar trend was observed in murine renal cancer cells (RENCA) (Supplementary Fig-1C). Furthermore, XL-184 treatment also led to an increase in the surface expression of both c-Met and AXL, as assessed by flow cytometry (Fig-1E), and this increase was further confirmed through confocal microscopy, which revealed increased cellular expression of both the proteins (Fig-1F). Notably, no significant changes in mRNA expression levels of c-Met or AXL was observed following XL-184 treatment (Fig-1G).

We then investigated whether the increased total c-Met levels observed following XL-184 treatment were associated with elevated receptor phosphorylation in the presence of its ligand, HGF. We found that pretreatment with XL-184 markedly blocked HGF-mediated c-Met phosphorylation only at concentrations of 100 nM and above. Interestingly, at lower Cabo concentrations (1 nM and 10 nM), we observed increased c-Met phosphorylation compared to HGF-treated controls in

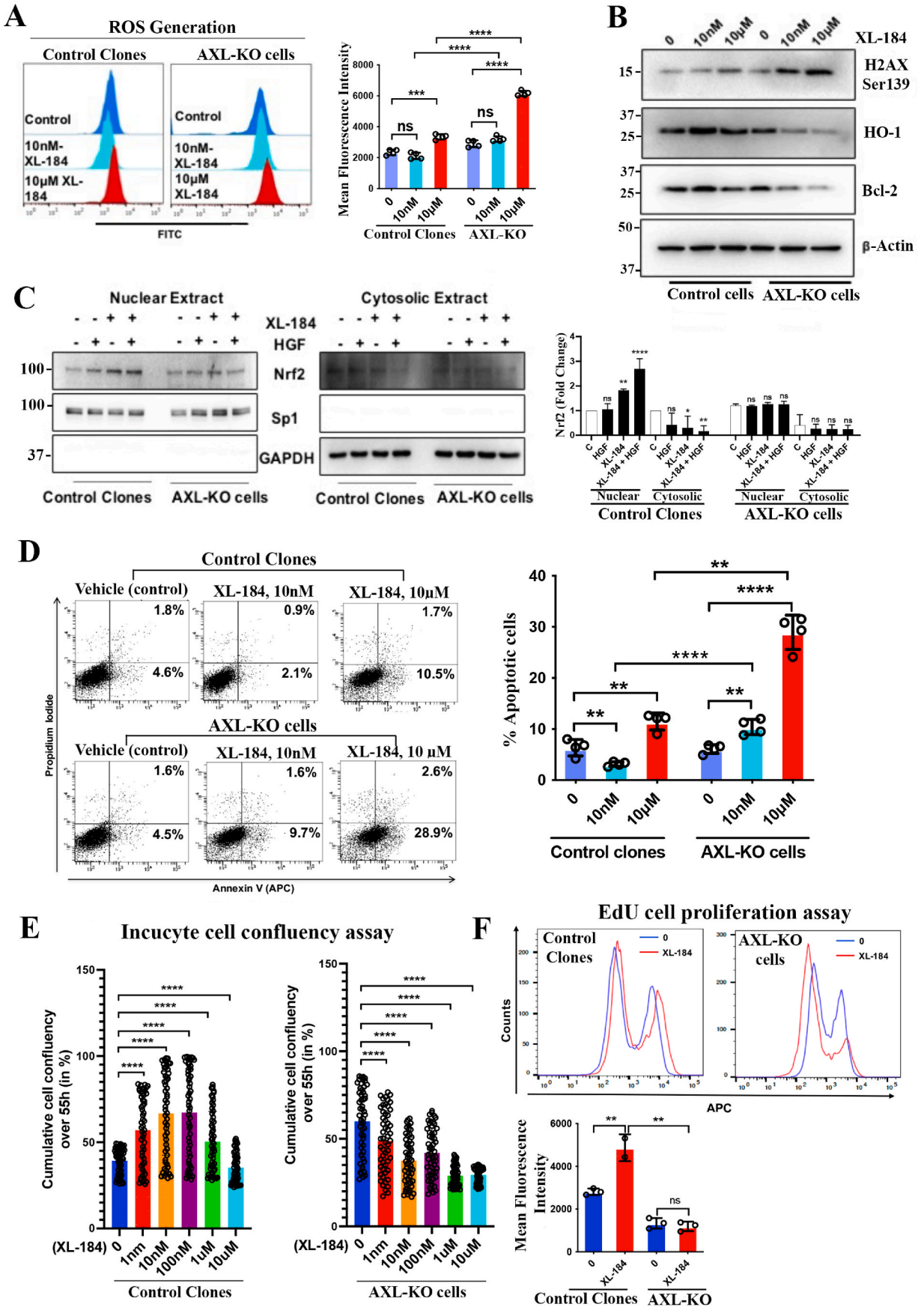

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Fig. 4. AXL-silenced renal cancer cells are more sensitive to XL-184 treat**ments. (A)** Control and AXL-KO cells were treated with low and high concentrations of XL-184 (10 nM & 10 μM) or vehicle for 6 h. Following treatments, cells were stained with ROS-ID dye, and total ROS generation was measured by flow cytometry. Quantifications are presented as bar graphs (right panel). Data represent the mean ± SD of four readings, representative of three different experiments. ***$p < 0.001$, ****$p < 0.0001$. **(B)** Control clones and AXL-KO cells were treated similarly as described in (A) for 24 h. Following treatments, the expression of phospho-H2AX (Ser139), HO-1, Bcl2, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments/biological replicates. **(C)** Control and AXL-KO cells were pre-treated with or without XL-184 (10 nM) for 12 h. Cells were then treated with or without HGF (50 ng/mL) for 12 h as indicated. Following treatments, isolation of nuclear and cytoplasmic fractions was performed as mentioned in Materials and Methods section, followed by Western blot to check the level of Nrf2. The purities of nuclear and cytoplasmic fractions were examined through the expression of Sp1 and GAPDH, respectively. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments. The bar graphs (*right panel*) represent the fold changes in Nrf2 expression by densitometry where the signals were normalized to internal controls Sp1/GAPDH. The columns represent the mean ±SD of readings of three different samples. ****$p < 0.0001$ and **$p < 0.01$ compared with controls. **(D)** Control clones and AXL-KO clones were treated similarly as described in (A) for 24 h. Cells were then stained with Annexin V (APC) and propidium iodide (PI), and apoptotic indices were analyzed by flow cytometry. The percentage of total (early + late) apoptotic cells was quantified and represented as bar graphs (right panel). Data represent the mean ± SD of four readings, representative of three different experiments. **$p < 0.01$, ****$p < 0.0001$. **(E)** Control and AXL-KO cells were seeded in 96-well plates in complete media containing 10 % FBS and treated with different concentrations of XL-184 (1 nM–10 μM). Plates were then subjected to live cell imaging under regular incubator conditions under the Incucyte live imaging system. Kinetic data of cell count per well over time (1-h intervals for 53 h) were plotted for each indicated concentration of XL-184 or vehicle. Data shown are representative of three independent experiments. The data points and error bars represent the mean ± SD of six technical replicates ($n = 6$). ****$p < 0.0001$. **(F)** Control and AXL-KO cells were cultured in complete media containing 10 % FBS and treated with XL-184 (10 nM) for 24 h. Following treatments, cells were collected and stained with Click-iT™ EdU Cell Proliferation assay reagent as mentioned in Materials and Methods and analyzed by flow cytometry. Quantifications are presented as bar graphs (bottom panel). Data represent the mean ± SD of three readings, representative of three different experiments. **$p < 0.01$.

all three renal cancer cell lines (786-O, ACHN and Caki-1) (Fig-1H).

To further validate these findings, we tested two additional c-Met inhibitors, crizotinib and PF-04217903. Both inhibitors also markedly increased the levels of total c-Met and AXL in RCC cells (Supplementary Fig-2A and 2B). Consistent with XL-184, crizotinib treatment blocked HGF-induced c-Met phosphorylation at concentrations of 100 nM and above, while lower concentrations of crizotinib promoted increased c-Met phosphorylation (Supplementary Fig-2C). PF-04217903 was found to be less effective, inhibiting HGF-induced c-Met phosphorylation only at concentrations of 1 μM and above (Supplementary Fig-2D).

Several reports suggest that the tumor microenvironment is characterized by significant gradients of oxygen, nutrients, and drugs, all of which can profoundly influence tumor growth and treatment responses [51,52]. Based on our observations, we hypothesize that the concentration-dependent inhibition and upregulation of the target receptor (c-Met) may lead to mixed outcomes. While initial tumor shrinkage may occur, areas of the tumor exposed to lower drug concentrations might experience increased c-Met expression (Fig-1I). Furthermore, the free availability of HGF in the system can trigger c-Met phosphorylation, activating signaling pathways that ultimately drive tumor progression and therapeutic resistance in later stages of the treatment.

3.2. *A lower concentration of cabozantinib/XL-184 promotes renal tumor growth in vivo, associated with increased c-met and AXL expression*

To investigate the effects of low doses of Cabo/XL-184 *in vivo*, we employed a renal tumor xenograft model. Interestingly, mice treated with XL-184 exhibited significantly larger tumor volumes compared to the vehicle treated control group (Fig-2A). Immunohistochemical analyses of the tumor tissues revealed that XL-184 treatment markedly increased the expression of c-Met, AXL, HO-1 and CD-31 (Fig-2B); and also, there was an increase in c-Met phosphorylation (Supplementary Fig-2E). We have also confirmed the expression levels of c-Met, AXL and HO-1 in tumor tissues by Western blotting (Supplementary Fig-2F). These findings suggest that prolonged treatment with a lower dose of XL-184 not only induces the expression of c-Met and AXL, but also promotes tumor growth.

Next, we evaluated the expression of c-Met and AXL in tumor tissues from renal cancer patients, both before and after XL-184 treatment. Similar to our observations in the pre-clinical model as well as *in vitro* studies, we observed a markedly higher expression of both c-Met and AXL in the tumor tissues from renal cancer patients following XL-184 treatment compared with untreated controls (Fig-2C); and the level of c-Met phosphorylation was also increased (Supplementary Fig- 2G). Together, these findings suggest that Cabo-mediated increased expression of c-Met and AXL could contribute to therapeutic resistance in renal cancer.

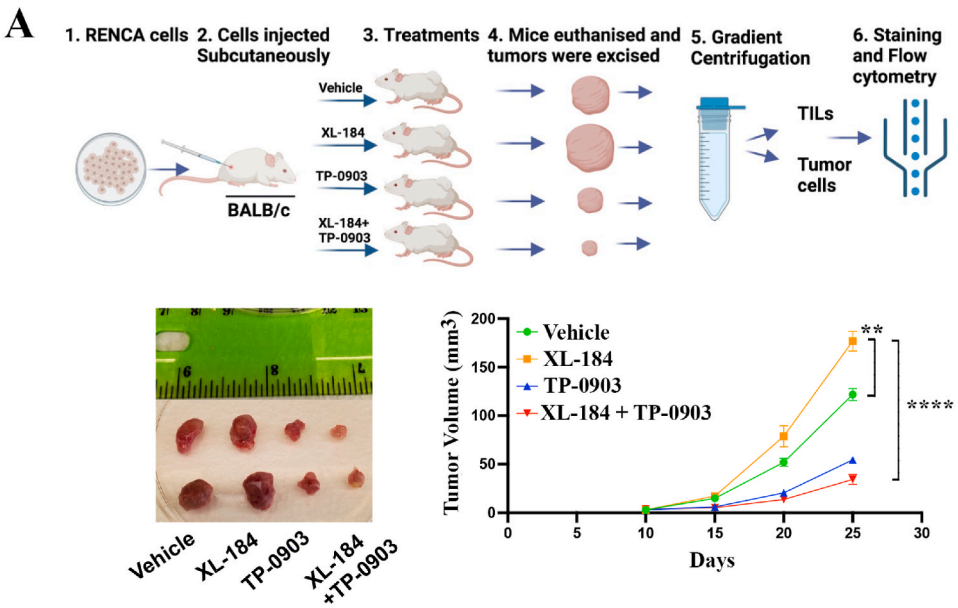
3.3. *Silencing of AXL prevents cabozantinib/XL-184-mediated c-met overexpression and its downstream signaling*

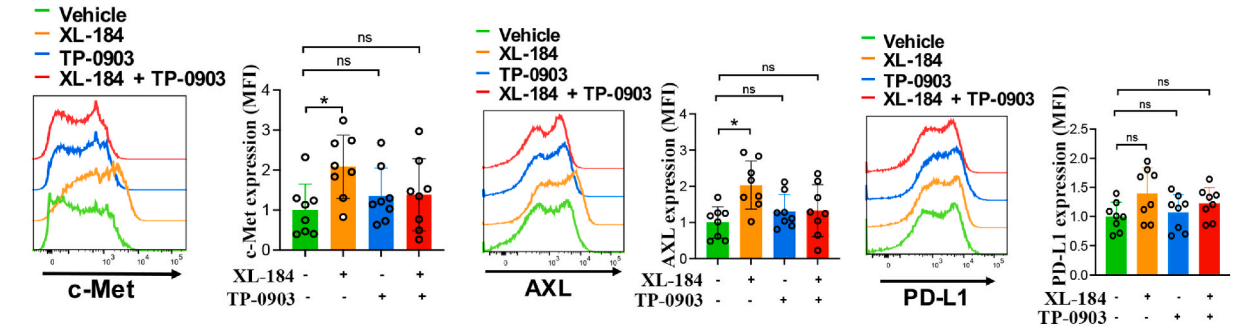
To study the role of AXL in Cabo-/XL-184-mediated c-Met over-expression, we utilized CRISPR-Cas9-mediated AXL-Knock out (KO) RCC cells (786-O) [23]. As expected, we found that in control clones, XL-184 treatment increased the expression of c-Met and AXL, consistent with previous observations (Fig-1D). However, in AXL-KO cells, treatment with XL-184 failed to increase c-Met expression (Fig-3A). Similar results were observed with crizotinib and PF-04217903 treatments in control and AXL-KO cells (Supplementary Fig-3A and 3B).

We also investigated the cell surface expression of both c-Met and AXL. Consistent with the increased total protein levels observed in Western blot analysis, XL-184 treatment significantly induced the surface expression of both c-Met and AXL in control clones. However, in AXL-KO cells, XL-184-induced increase in c-Met surface level was significantly prevented (Fig-3B). To further validate the role of AXL in XL-184-mediated c-Met over-expression, we used TP-0903, a small molecule inhibitor of AXL. Treatment with TP-0903 effectively prevented XL-184-mediated increase c-Met expression in both human (786-O, ACHN and Caki-1) and mouse (RENCA) renal cancer cells compared to controls (Fig-3C). Furthermore, we confirmed that TP-0903 reduces HGF-mediated c-Met phosphorylation and downstream signaling events (Supplementary Fig-4A and 4B).

We next explored how the XL-184 treatment (at a concentration of 10 nM) modulates ligand (HGF)-induced c-Met phosphorylation and its downstream signaling pathways (Erk1/2, Akt and Stat3) in RCC cells (786-O and ACHN). As anticipated, HGF treatment phosphorylated c-Met (tyrosine-1234/1235 and −1349) and induced the activation of pErk1/2, pAkt and pStat3 (Fig-3D). Importantly, XL-184-mediated overexpression of c-Met led to enhanced phosphorylation of the receptor compared with controls, and it also amplified the phosphorylation of Erk1/2, Akt and Stat3 in both 786-O and ACHN cells (Fig-3D). In contrast, when the cells were pretreated with a combination of XL-184 and TP-0903, HGF-induced c-Met phosphorylation, along with the activation of Erk1/2, Akt and Stat3 was significantly inhibited (Fig-3D).

Together, our data suggest that Cabo/XL-184-mediated increased c-Met expression enhances its phosphorylation in the presence of HGF, which in turn amplifies downstream signaling pathways that can promote tumor survival and this may contribute to therapeutic resistance. Notably, combination treatment of XL-184 and TP-0903 not only





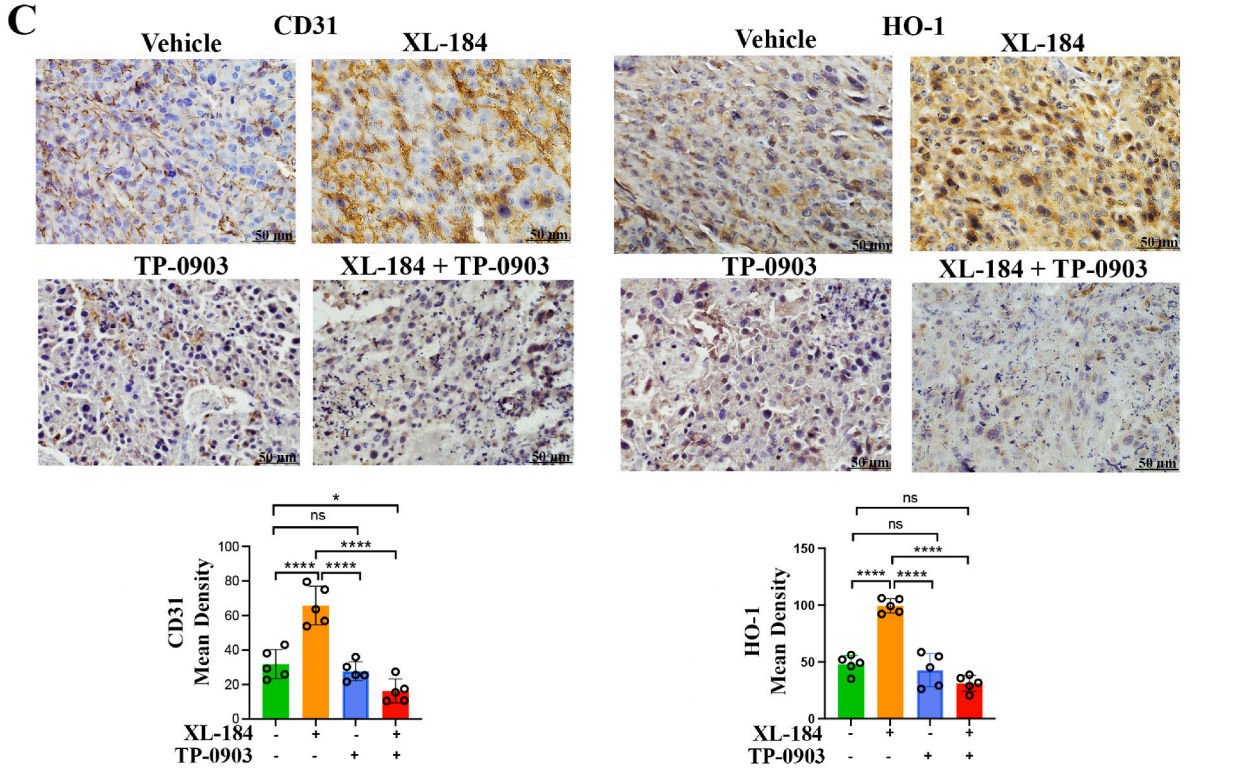


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Fig. 5. A combination of XL-184 and TP-0903 is more effective in restricting tumor growth in a syngeneic Balb/c mice **model. (A)** RENCA cells were injected into the flanks of Balb/c mice and treated with either vehicle, XL-184 (7.5 mg/kg/day, IP), TP-0903 (10 mg/kg/day, IP), or the combination for three weeks (n = 5 per group). Tumor volumes were measured. The line graph shows tumor volumes recorded at the indicated time points (bottom right panel). **p < 0.01 and ****p < 0.0001. **(B)** Tumors were dissociated into single-cell suspensions as described in the Materials and Methods section and stained for extracellular c-Met, AXL, and PD-L1 expression by flow cytometry. Representative of three experiments. *p < 0.05. **(C)** Renal tumor sections obtained from (A) were processed for immunohistochemistry. The panel shows representative photomicrographs (magnification × 400) showing immunohistochemical staining (brown) of CD31 and HO-1 in harvested tumor tissues. Columns represent the average intensity values of stained proteins; bars represent ±SD of five different readings. *p < 0.05, ****p < 0.0001. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

inhibits c-Met activation but also suppresses its downstream signaling pathways, potentially offering a therapeutic strategy to counteract resistance.

3.4. Knock-out of AXL facilitates Cabo/XL-184-mediated increased oxidative stress and apoptosis of renal cancer cells and effectively inhibits cell proliferation

Chemotherapeutic agents often induce the generation of reactive oxygen species (ROS) to promote apoptosis. In this study, we investigated the effect of XL-184 on ROS generation in control and AXL-KO renal cancer cells. We found that, in AXL-KO RCC cells, XL-184 treatment markedly increased ROS generation compared with control cells (Fig. 4A).

We further analyzed the markers modified during oxidative stress. Phosphorylation of gamma H2AX at Ser 139 is a key marker of DNA damage. In control cells, H2AX phosphorylation was slightly increased following XL-184 treatment. However, in AXL-KO clones of RCC cells, XL-184 treatment markedly increased the phosphorylation of gamma H2AX (Fig-4B). Additionally, compared to control cells, XL-184 treatment markedly decreased the expression of redox-sensitive and anti-apoptotic proteins such as HO-1 and Bcl-2 in AXL-KO cells.

As previously mentioned, the transcription factor Nrf2 is activated following c-Met activation and plays a major role in relieving oxidative stresses in tumor cells [19]. Upon activation, Nrf2 translocate to the nucleus. In control cells, HGF-induced nuclear translocation of Nrf2 was not inhibited following XL-184 treatment. However, in AXL-KO RCC cells, XL-184 treatment markedly restricted HGF-induced nuclear translocation of Nrf2 (Fig-4C). The expression of Sp1 and GAPDH was used to evaluate the purity of nuclear and cytoplasmic fraction, respectively.

Next, we studied the effect of XL-184 on apoptosis in both control and AXL-KO cells under similar treatment conditions. As shown in Fig-4D, the lower dose (10 nM) of XL-184 reduced the total apoptotic cells (early + late) from 6.3 % (vehicle-treated) to 3.1 % in control clones, while the higher dose (10 μM) of XL-184 increased the total apoptotic cells to 11.4 %. In contrast, in AXL-KO cells, the lower dose of XL-184 significantly increased total apoptotic cells from 6.1 % (vehicle-treated) to 10.4 %. Moreover, the higher dose of XL-184 further increased the total apoptotic cells to 28.9 %. Similar findings of increased renal cancer cell apoptosis were observed with a combination treatment of XL-184 and TP-0903 (Supplementary Fig-5A-C).

We also examined the effects of increasing concentrations (1 nM - 10 μM) of XL-184 on cell proliferation using Incucyte automated live imaging system. In control clones, lower concentrations of XL-184 (1 nM–100 nM), significantly increased cell proliferation in a time dependent manner. However, in AXL-KO cells, there was a significant reduction in cell proliferation (Fig. 4E). Finally, we evaluated the effect of XL-184 on DNA replication of RCC cells. Our data indicate that the lower concentration of XL-184 significantly increased DNA replication in control clones but no significant change was observed in AXL-KO cells (Fig-4F).

Together, our results suggest that the induction of AXL can promote therapeutic resistance against c-Met/RTK inhibitors like Cabo/XL-184, particularly at lower concentrations, by decreasing oxidative stress through Nrf2 activation. This results in reduced apoptosis and enhanced proliferation of renal cancer cells. The knock-out of AXL, however, markedly facilitates cancer cell death following XL-184 treatment.

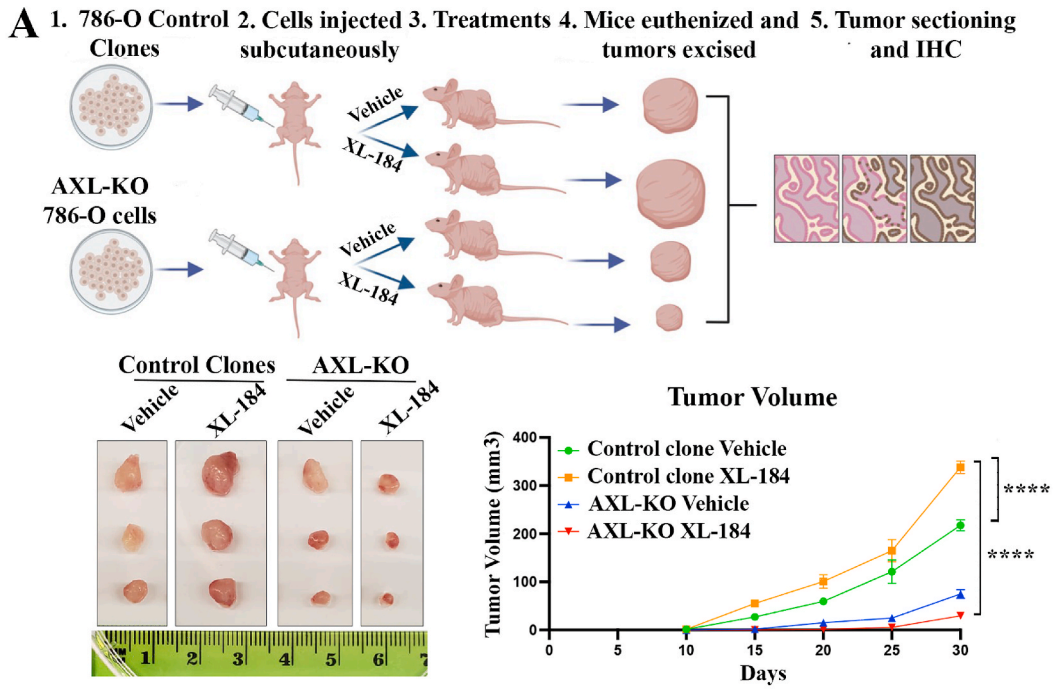
3.5. A combination treatment with Cabo/XL-184 and TP-0903 significantly reduces renal tumor growth in a syngeneic murine model

To evaluate the physiological relevance of our *in vitro* findings, we investigated the effect of the combination of XL-184 and TP-0903 on renal tumor growth. Using an immunocompetent Balb/c mouse model, we injected syngeneic renal cancer cells, into the mice and treated them with either a combination of XL-184 (at a lower concentration) and TP-0903 or the vehicle alone (control). As shown in Fig-5A, XL-184 treatment alone resulted in a marked increase in renal tumor volume compared to controls, while the combination treatment (XL-184 + TP-0903) markedly reduced the tumor growth.

To monitor the efficacy of XL-184 + TP-0903 combination treatment *in vivo*, we dissociated the tumors into single cell suspensions and stained tumor cells for extracellular c-Met, AXL and PD-L1 expression. We observed an increase in c-Met and AXL levels in XL-184-treated mice, but the combination treatment led to a marked decrease in the expression of both c-Met and AXL in tumor cells (Fig-5B). Additionally, while there was a non-significant trend for increased PD-L1 expression in tumor cells from XL-184-treated mice, the combination treatment resulted in decreased PD-L1 expression. We also assessed tumor vessel density (CD31) and oxidative stress (HO-1) in the tumor tissues. Although XL-184 treatment alone increased both CD31 and HO-1 expression, the combination treatment with TP-0903 markedly reduced their expression (Fig-5C). We also confirmed our findings through Western blot analyses of tumor tissue extracts, where it showed that the combination treatment downregulated the expression of c-Met, AXL and HO-1 (Supplementary Fig-6A). These findings suggest that XL-184 + TP-09003 combination treatment is effective *in vivo*, and might influence AXL-c-Met signaling, potentially modulating the immune evasion or the immunogenicity of tumor cells.

To explore this possibility, we performed immunohistochemistry of the renal tumor tissues to check the infiltration of immune cells. Our results suggested that the combination treatment (XL-184 + TP-0903) increased the infiltration of $CD4^+$, $CD8^+$ as well as CD11c + cells within tumor tissues compared with vehicle-treated control (Supplementary Fig- 6B). We also isolated tumor infiltrating lymphocytes (TILs) from excised mouse renal tumors and analyzed immune infiltration via flow cytometry. We did not observe significant differences in the infiltration of $CD8^+$ T cells, frequencies of exhausted PD1+ TIM3+ $CD8^+$ T cells, or $CD4^+$ FOX3+ regulatory T cells across the four experimental groups. However, we did note a non-significant trend for increased activation of $CD69^+$ $CD8^+$ T cells in tumors from mice treated with the XL-184 + TP-0903 combination, compared to control groups (Supplementary Fig-7A-D).

Together, these *in vivo* results support our *in vitro* findings, suggesting that, similar to renal cancer cell lines, the lower dose of Cabo/XL-184 increases the expression of both c-Met and AXL in tumor tissues, potentially promoting enhanced tumor growth. The inhibition of AXL through TP-0903 treatment markedly reduced tumor growth by decreasing HO-1 expression, thereby increasing oxidative stress and facilitating tumor cell death. However, there were no significant changes in the infiltration patterns of TILs following the combination treatment.



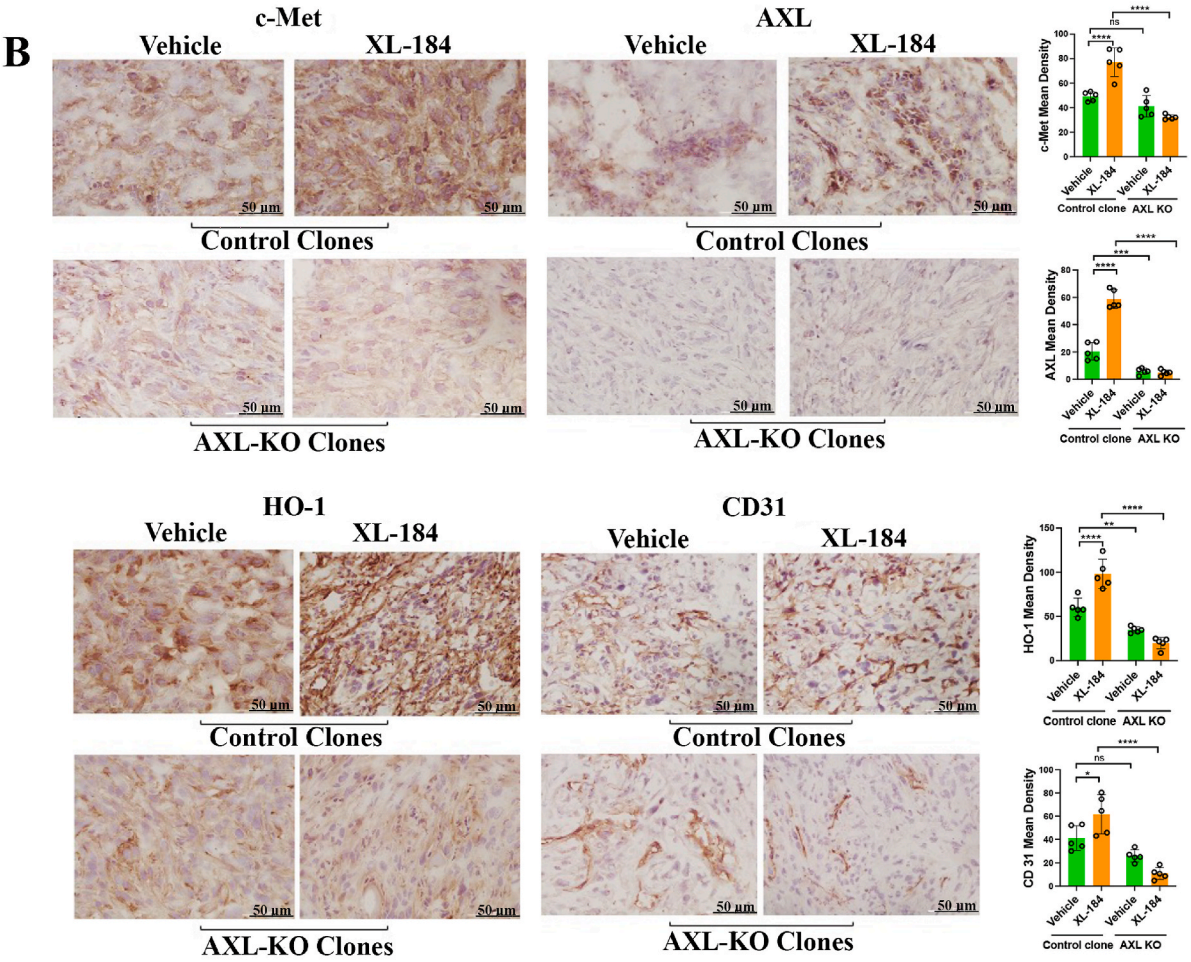


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Fig. 6. XL-184 is more effective in restricting tumor growth in AXL-KO cells in a tumor xenograft **model. (A)** Schematic representation of the experiment. Control clones and AXL-KO cells were subcutaneously injected into the flanks of athymic nude mice. Once palpable tumors formed, mice were injected intraperitoneally (IP) with XL-184 (7.5 mg/kg) or vehicle (n = 5 per group) for 3 weeks, and tumor volumes were measured. Representative photographs of tumors excised on the 25th day following tumor injection (bottom left panel). The line graph shows tumor volumes recorded at the indicated time points (bottom right panel). ****$p < 0.0001$. **(B)** Renal tumor sections were processed for immunohistochemistry. The figure illustrates representative photomicrographs (magnification × 400) showing immunohistochemical expression (brown) of c-Met, AXL, CD31, and HO-1 in harvested tumor tissues. Columns represent the average intensity values of stained proteins; bars represent ±SD of five different readings. *$p < 0.05$, **$p < 0.01$ ***$p < 0.001$, and ****$p < 0.0001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.6. Cabo/XL-184 significantly reduces renal tumor growth when AXL is silenced in a tumor xenograft model

To further validate our findings that AXL is essential and plays a significant role in XL-184-mediated increased renal tumor growth, we used a tumor xenograft model in immunodeficient nude mice. Mice were injected with either CRISPR-Cas9-mediated AXL-KO or control renal cancer cells (786-O) and treated with XL-184 or vehicle as described (Fig-6A). In contrast to our *in vivo* findings (Fig-2) and consistent with our *in vitro* observations (Fig-4), XL-184 treatment markedly reduced renal tumor volume in AXL-KO cells compared to control tumors (Fig-6A). Furthermore, XL-184-mediated increased expression of c-Met, CD31 and HO-1 in control renal tumors were significantly restricted in AXL-KO tumors (Fig. 6B). These results suggest that AXL plays a critical role in promoting c-Met-induced renal tumor growth following treatment with a lower concentration of XL-184.

3.7. Cabo/XL-184 treatment stabilizes c-Met in renal cancer cells

To understand the mechanistic role(s) of AXL in regulating c-Met, the control and AXL-KO renal cancer cells were treated with HGF, and time kinetics (ranging from 15 min to 6 h) was performed. We found that in control clones, c-Met remained strongly phosphorylated for an extended time-period, beyond 30 min. However, in AXL-KO cells, HGF-mediated c-Met phosphorylation was short-lived and started to deplete after 30 min (Fig. 7A).

To investigate the effect of XL-184 on c-Met phosphorylation and degradation, RCC cells were treated with or without a lower concentration of XL-184 (10 nM). As shown in the Fig-7B, at this concentration, XL-184 did not block HGF-mediated c-Met phosphorylation; rather, it increased the total c-Met levels and c-Met phosphorylation. In the absence of XL-184 pre-treatment, HGF-mediated c-Met phosphorylation and total c-Met levels began to decrease after 1–3 h. These data suggest that XL-184 can extend the half-life and accumulation of c-Met, allowing it to remain phosphorylated in the presence of HGF.

To further explore the role of c-Met and AXL crosstalk, RCC cells were treated with a lower concentration (10 nM) of XL-184, and an immunoprecipitation assay was performed. We found that XL-184 treatment increased the association of c-Met and AXL, likely stabilizing the c-Met/AXL complex. This explains why XL-184 treatment does not increase c-Met expression in the absence of AXL (Fig-7C).

Next, we investigated the effect of HGF-mediated c-Met internalization and the impact of XL-184 on this process. As shown in Fig. 7D, HGF treatment (1–3 h) caused a decrease in surface levels of c-Met in RCC cells. However, cells pretreated with XL-184 blocked the HGF-mediated c-Met internalization, further indicating that XL-184 can interfere with the negative regulation of c-Met, thereby contributing to its increased levels (Fig-7D).

As mentioned in the earlier section, the phosphorylation at Y1003 is critical for the degradation and regulation of c-Met [53,54]. The phosphorylation at Y1003 creates a docking site for E3 ligase binding, leading to receptor ubiquitination and subsequent degradation [10,55]. To understand the mechanism behind XL-184-mediated increased levels of total c-Met, we analyzed how XL-184 treatment (at an effective inhibitory concentration) can modulate HGF-induced c-Met phosphorylation at tyrosine Y1003 residue. As shown in Fig-7E, in both 786-O and ACHN cells, XL-184 treatment completely blocked the phosphorylation of c-Met at Y1003; and this might lead to decreased ubiquitination of c-Met, which can allow increased accumulation of the receptor. To confirm our finding, through immunoprecipitation studies, we found that HGF treatment increased the ubiquitination of c-Met, whereas XL-184 treatment indeed inhibited c-Met ubiquitination (Fig. 7F).

It has been shown that after ubiquitination, c-Met is degraded in either proteasomes or lysosomes [10,14,56]. To investigate if XL-184 treatment can modulate these processes, RCC cells were treated with a combination of MG-132 (a proteasomal inhibitor) and XL-184. As shown in Fig-7G, inhibiting proteasomes resulted in increased c-Met levels, and the combination treatment further elevated total c-Met levels in both 786-O and ACHN cells.

Together, our findings suggest that Cabo/XL-184 treatment promotes the accumulation of c-Met within RCC cells by decreasing the phosphorylation at Y1003, thereby inhibiting its ubiquitination and degradation, and by increasing its association with AXL.

3.8. Silencing AXL reverses therapeutic resistance to Cabo/XL-184 in renal cancer cells

To investigate mechanisms of resistance to Cabo/XL-184, we generated Cabo-resistant renal cancer cells (Caki-1R/CaboR) following prolonged treatment, as described. To confirm resistance, we performed an apoptosis assay in both Caki-1S (control/Cabo sensitive/CaboS) and Cabo-resistant (Caki-1R) cells, treated with increasing concentrations of XL-184 (5–20 μM). As shown in Supplementary Fig-8A, XL-184 significantly induced apoptosis (early + late apoptotic cells) in control Caki-1S cells at 10 μM and 20 μM. However, Caki-1R cells exhibited no significant increase in apoptosis following XL-184 treatments, confirming resistance.

Epigenomic profiling revealed marked differences between Cabo-resistant (Caki-1R/CaboR) and Cabo-sensitive cells (Caki-1S/CaboS). H3K27ac ChIP-seq analysis identified differentially acetylated regions in CaboR versus CaboS cells (Figure-8A), with increased activation of transcription factors (TFs) such as ETV1, HOXA9, and FOXK1, in resistant cells. Additionally, CaboR cells exhibited upregulation of genes involved in oxidative phosphorylation, cell cycle progression, DNA replication, DNA repair, and the reactive oxygen species (ROS) pathway, alongside enrichment of NRF2 pathway and PTEN regulation gene sets (Fig. 8B and C). Conversely, genes associated with glycolysis, apoptosis, hypoxia response were downregulated in CaboR cells. However, based on our RNA seq-based gene set enrichment analysis (GSEA) results (not shown), we did not observe any significant enrichment of other pathways, such as EGFR, HER2 or VEGFR2 in resistant cells.

Further analysis of molecular markers revealed significantly elevated expression of AXL, c-Met, Bcl-2, HO-1 and Rad51 in Cabo resistant cells (CaboR/Caki-1R) compared to Cabo-sensitive cells (Cabo-1S/CaboS) (Supplementary Fig-8B), suggesting possible roles of these molecules in the development of resistance.

To assess the role of AXL in XL-184 resistance, we treated Caki-1R cells with TP-0903 (AXL-specific inhibitor) and performed an apoptosis assay. As shown in Fig-8D (*left panel*), TP-0903 significantly induced apoptosis in XL-184 resistant cells, increasing apoptotic cells (early + late) from 4.3 % (vehicle-treated cells) to 19.3 % (TP-0903-treated cells). TP-0903 also downregulated AXL, c-Met, Bcl-2, HO-1 and Rad51 expression (Fig-8D, *right panel*). In addition, to confirm the role of ETV1, we found that the treatment of resistant RCC cells with YK-4-279

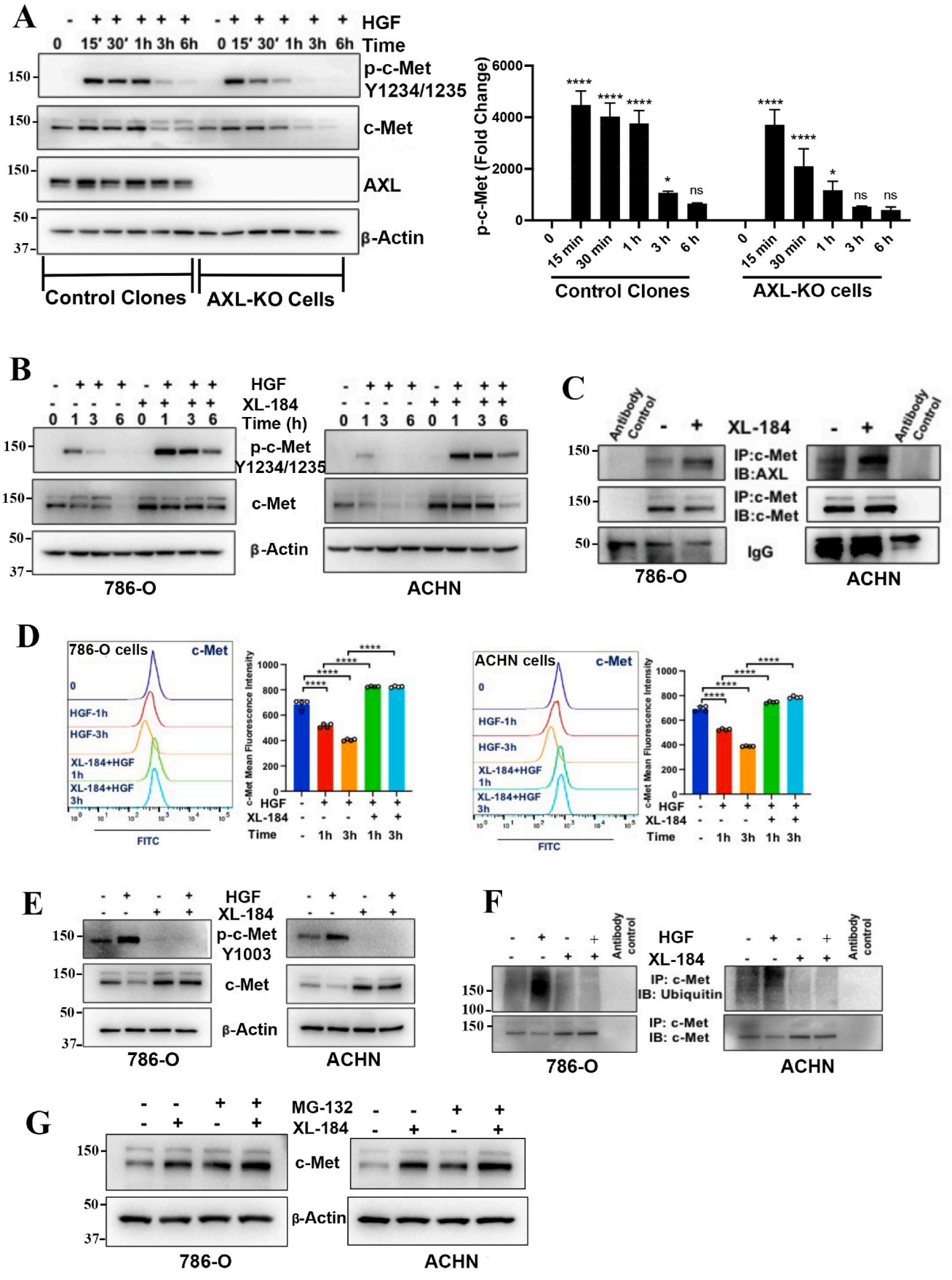

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Fig. 7. XL-184 treatment increases c-Met expression by interfering with the degradation of **c-Met. (A)** Serum-starved control and AXL-KO cells were treated with or without HGF (50 ng/mL) for the indicated time points. Following treatments, the expression of phospho-c-Met (Tyr1234/1235), c-Met, AXL, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments. The bar graphs represent the fold changes in phospho-c-Met expression by densitometry where the signals were normalized to an internal control β-actin. The columns represent the mean±SD of readings of three different samples. ****$p < 0.0001$ and *$p < 0.05$ compared with controls. **(B)** 786-O and ACHN cells were treated with or without XL-184 (10 nM). After 24 h, cells were treated with or without HGF (50 ng/mL) for the indicated time points, and the expression of phospho-c-Met (Tyr1234/1235), c-Met, and β-actin in cell lysates was analyzed by Western blot. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments/biological replicates. **(C)** 786-O and ACHN cells were treated with or without Xl-184 (10 nM) for 24h. Following treatment cell lysates were subjected to immunoprecipitation (IP) assay using anti-c-Met, and blotted (IB) with either anti-AXL or anti-c-Met antibodies. Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments/biological replicates. **(D)** 786-O and ACHN cells were treated with HGF (50 ng/mL) alone for the indicated time points or in combination with pre-treatment (5 min) with XL-184 (1 μM). Following treatments, the extracellular level of c-Met was analyzed by flow cytometry, and the quantifications are presented as bar graphs. Data represent the mean ± SD of four readings, representative of three different experiments. ****$p < 0.0001$. **(E)** Serum-starved 786-O and ACHN cells were pre-treated with XL-184 (1 μM) for 2 h, followed by treatment with or without HGF (50 ng/mL) for 1 h. The expression of phospho-c-Met (Tyr1003), c-Met, and β-actin in cell lysates was analyzed by Western blot. **(F)** 786-O and ACHN cells were treated similarly as in (D). Following treatments, cell lysates were subjected to immunoprecipitation (IP) assay using anti-c-Met antibodies, and blotted (IB) with either anti-ubiquitin or anti-c-Met antibodies. **(G)** 786-O and ACHN cells were treated with combinations of MG132 (0.5 μM) and XL-184 (1 μM) for 18 h. Following treatments, the expression of c-Met, and β-actin in cell lysates was analyzed by Western blot. E-G, Molecular weight markers (kDa) are indicated at the left. Representative of three independent experiments/biological replicates.

(an ETV1 inhibitor) significantly promoted cancer cell apoptosis (Supplementary Fig- 9).

To further validate the role of AXL in resistance, we silenced AXL in Cabo-resistant cells (Caki-1R) using siRNA and performed an apoptosis assay. As shown in Fig-8E (*left panel*), AXL knockdown significantly inducted apoptosis in resistant cells compared to controls. Additionally, AXL, c-Met, Bcl-2, HO-1 and Rad51 expression were markedly reduced in AXL-siRNA-transfected Caki-1R cells (Fig-8E, *right panel*).

These findings indicate that AXL plays a crucial role in mediating resistance to Cabo/XL-184. Targeting AXL, either through pharmacological inhibitor (TP-0903) or gene silencing (siRNA), effectively induce apoptosis in resistant cells, which exhibit distinct epigenomic and transcriptomic profile. These results highlight AXL as a potential therapeutic target to overcome resistance in Cabo-resistant renal cancer.

4. Discussion

Current treatment options for the renal cell carcinoma (RCC) patients primarily include tyrosine kinase inhibitors (TKIs), anti-angiogenic agents, immunotherapy and combination therapies [57,58]. However, most patients eventually develop therapeutic resistance [59]. Cabozantinib (Cabo/XL-184), a c-Met/RTK inhibitor, is approved for the treatment of advanced RCC, yet acquired drug resistance remains a significant clinical challenge. Additionally, Cabo is frequently combined with immunotherapy, but the toxicity of these regimens limits their clinical utility [60,61]. Given that Cabo is one of the most effective tyrosine kinase inhibitors (TKIs) for RCC, it is crucial to identify strategies to enhance its therapeutic efficacy and overcome resistance. A previous study showed that targeting c-Met and AXL, another RTK, can overcome the acquired resistance against sunitinib, a VEGFR inhibitor [62]. Our study also highlights the potential of targeting AXL, which is overexpressed in RCC, to mitigate resistance against c-Met inhibitors.

We found that c-Met and AXL form a complex, and treatment with lower concentrations of Cabo/XL-184, resulted in increased expression of both total c-Met and AXL in RCC cells. Interestingly, while higher concentrations (100 nM and above) of Cabo suppressed c-Met phosphorylation, lower doses (1 nM and 10 nM) paradoxically increased c-Met phosphorylation. For optimal pro-apoptotic functions of anti-tumor agents, they must be distributed throughout the tumor [63]. However, in most cases, the drugs are heterogeneously distributed within the tumor; and thus, only limited portions of the tumor are exposed to the required concentration(s) of therapeutic agents [64,65]. Thus, the cancer cells that are sensitive to anti-cancer agents in tissue culture may become resistant when growing as a tumor. Our *in vivo* xenograft and syngeneic models confirmed that prolonged treatment with lower Cabo doses led to increased levels of c-Met/AXL expression, c-Met phosphorylation and tumor growth. Furthermore, tumor tissues from Cabo-treated renal cancer patients exhibited higher levels of c-Met/AXL and c-Met phosphorylation. Clinical evidence further supports the association between AXL expression and response to Cabo. A recent analysis of pretreatment tumor tissues from a cohort of 278 patients with metastatic RCC in phase III METEOR trial demonstrated that high AXL expression was associated with greater clinical benefit from Cabo (which can partly inhibit AXL) compared to everolimus [66]. This suggests that the antitumor activity of Cabo in RCC is mediated, in part, by AXL inhibition. In this study, our findings suggest that within the tumor microenvironment, suboptimal Cabo concentrations might promote tumor progression through increased interactions of c-Met and AXL. However, direct quantification of Cabo levels in different tumor regions based on c-Met/AXL expression can further strengthen the physiological significance of our findings. It has been reported that lower doses of Cabo are associated with greater risk of disease progression compared with the higher doses, supporting the hypothesis that insufficient Cabo exposure may drive resistance [67,68]. Our study suggests that silencing of AXL, either through genetic knockout or pharmacological inhibition, can prevent Cabo-induced c-Met overexpression, phosphorylation, and activation of downstream signaling events.

Most anticancer agents exert their effects by inducing apoptosis through increased ROS generation. However, cancer cells can counteract ROS-mediated apoptosis by activating the Nrf2 pathway and its effectors, such as HO-1 [69]. Our epigenomic analysis revealed that the Nrf2 pathway is markedly upregulated in Cabo-resistant RCC cells. We found that lower concentrations of Cabo induced Nrf2 activation and HO-1 overexpression, which were markedly reduced following AXL knockout. Importantly, Cabo-induced apoptosis was significantly enhanced in AXL-KO cells. Our previous studies [19,69] demonstrated that c-Met activation can induce Nrf2 in RCC cells. Here, our data suggest that AXL plays a pivotal role in promoting RCC cell survival and therapeutic resistance by mitigating Cabo-induced oxidative stress via the c-Met-Nrf2-HO-1 pathway. Targeting AXL could therefore be a promising strategy to overcome resistance and promote apoptotic cell death. In support of our findings, studies in small-cell lung cancer have shown that inhibiting AXL can overcome resistance against to DNA repair and cell-cycle checkpoint inhibitors [70]. Furthermore, TP-0903, an AXL inhibitor, has demonstrated clinical safety and efficacy, highlighting its potential therapeutic relevance [71]. One of the major Nrf2-regulated genes is SLC7A11 [72]; and it has been shown that in pancreatic cancer, SLC7A11 plays a major role in mediating therapeutic resistance against ferroptosis, another mechanism of cell death [73]. Although our findings suggest the critical role of the AXL-c-Met-Nrf2 pathway in mediating therapeutic resistance against apoptotic cell death in RCC, the possible role of ferroptosis (in addition to apoptosis) cannot be ruled out and needs to be further explored. In addition, the c-Met-Nrf2 pathway can also regulate immune-related molecules,

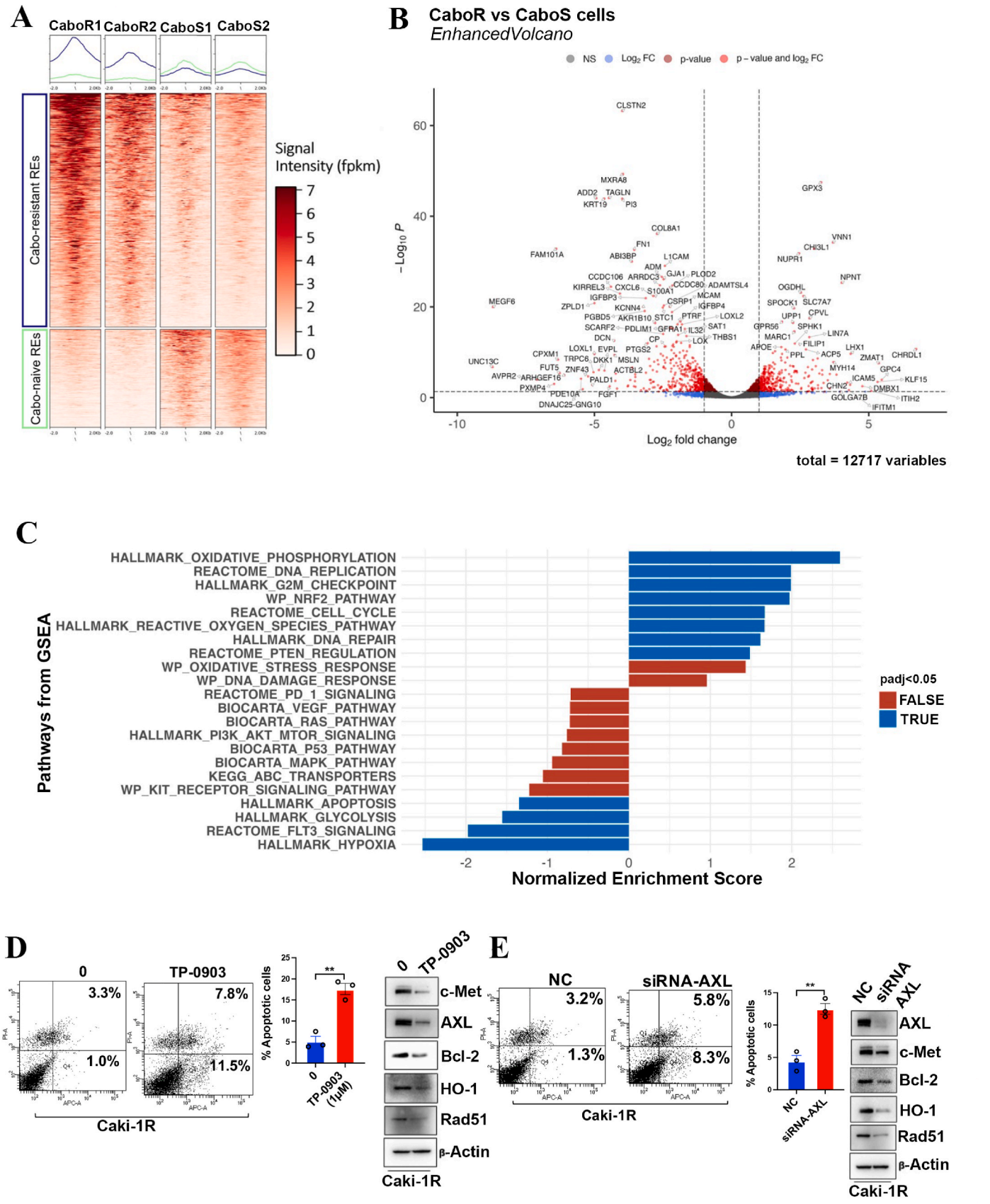

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Fig. 8. Cabo (XL-184)-resistant Caki-1 cells becomes sensitive to XL-184 treatment by targeting **AXL.** Caki-1 cells were cultured following the procedures outlined in the Materials and Methods section, while Caki-1R/CaboR cells were continuously cultured in the presence of XL-184 (10 μM). **(A)** Heatmap showing signal intensity at H3K27ac sites in cabozantinib-resistant cells (Caki-1R/CaboR) compared to wild-type cabozantinib-sensitive Caki-1 cells (CaboS). **(B)** Volcano plot of upregulated and downregulated genes in cabozantinib-resistant cells (Caki-1R/CaboR) compared to wild-type cells (CaboS). **(C)** Upregulated and downregulated pathways from cabozantinib-resistant cells (Caki-1R/CaboR) compared to wild-type cells. **(D)** Caki-1R cells were treated with or without (control) TP-0903 for 24 h. Subsequently, cells were stained with Annexin V (APC) and propidium iodide (PI). Apoptotic indices were then analyzed using flow cytometry, and the percentage of total (early + late) apoptotic cells was quantified and presented as bar graphs (*left panels*). Data represent the mean ± SD of triplicate readings and are representative of three independent experiments. $^{**}p < 0.01$. Following treatments, the expression levels of AXL, c-Met, Bcl-2, HO-1, Rad51, and β-actin in cell lysates were assessed via Western blot (*right panel*). **(E)** Caki-1R cells were transfected with NC-siRNA (control) and AXL-siRNA. Following transfections, an apoptosis assay was performed similarly as described in A. Data represent the mean ± SD of triplicate readings and are representative of three independent experiments. $^{**}p < 0.01$. Following transfection, the expression levels of AXL, c-Met, Bcl-2, HO-1, Rad51, and β-actin in cell lysates were assessed via Western blot (*right panel*).

including PD-L1 [74]; and it has been shown that in non-small cell lung cancer (NSCLC), targeting AXL can restore PD1 blockade sensitivity through the expansion of CD8 T cells [75]. The trends of our observations in this study also suggest that a combination therapy using Cabo + AXL inhibitor might be promising to promote immune-mediated cell death in RCC.

Our findings also indicate that c-Met-AXL complex formation is enhanced following Cabo treatment, which may contribute to acquired resistance. Similar to our observation, it has been shown that the RTK, EGFR, can also directly interact (cross-talk) with several cancer-relevant proteins, including AXL, EphA2, and CUB domain-containing protein-1 to develop resistance [76]. Several mechanisms can be involved in AXL-c-Met activation for mediating therapeutic resistance. It has been reported that heparan sulfation can play a major role in the activation of multiple RTKs, including AXL [77]. Notably, in this study, we show that silencing of AXL prevented Cabo-induced c-Met accumulation. Previous studies suggest that AXL promotes RCC invasion and metastasis by activating c-Met through Src [78]. Similar to RCC, AXL can also regulate the growth-promoting events in some other cancers to mediate tumor cell survival. It has been shown that AXL is a mediator of YAP-dependent oncogenic activities in hepatocellular carcinoma [79]. Induction of AXL can mediate resistance to EGFR-targeted therapy in lung cancer [80]. Co-activation of AXL and CDCP1 in NSCLC can limit the efficacy of EGFR TKIs [81]. Although Cabo can inhibit AXL at higher concentrations, it is not AXL-specific. We confirmed that the inhibition of AXL using TP-0903 mimics the effects as observed in AXL-KO cells. Cabo treatment appears to interfere with c-Met degradation. Different c-Met tyrosine residues regulate distinct functions [82]. Y1234/1235 and Y1349 promote tumor progression, whereas Y1003 is critical for c-Met ubiquitination and degradation [10–14]. Mutations at Y1003 impairs c-Met binding to Cbl (E3 ubiquitin ligase), leading to reduced ubiquitination and prolonged receptor stability [10,53]. Our study shows that Cabo treatment not only blocks Y1234/1235, but also inhibits Y1003 phosphorylation, suggesting a potential mechanism for c-Met stabilization; and it is possible that the observed effect is through the reduced binding of c-Met to Cbl, and it can be regulated by AXL. This stabilization may be further facilitated by c-Met-AXL interactions. Supporting our findings, previous studies have shown that reduced proteolytic shedding of receptor tyrosine kinases is a post-translational mechanism of kinase inhibitor resistance [83]. On the other hand, it is possible that Cabo-mediated increased expression of AXL might be an adaptive transcriptional response.

To further investigate the resistance mechanism(s), we generated Cabo-resistant RCC cells and found AXL inhibition significantly induced apoptosis in these cells. Epigenomic profiling revealed substantial alterations in histone modifications, particularly H3K27ac, suggesting that prolonged Cabo exposure derives major epigenetic changes that reshape transcriptional states and promote resistance. One key transcription factor, ETV1, can exhibit strong promoter and enhancer activation in Cabo-resistant cells. ETV1 regulates the PI3K/Akt and MAPK/Erk pathways, which are major downstream effectors of c-Met-AXL signaling, promoting cell survival, proliferation, and metastasis [84,85]. Our results demonstrate that combining TP-0903 and Cabo markedly reduced HGF-induced c-Met phosphorylation and downstream Erk1/2 and Akt activation confirming the role of AXL in resistance.

Additionally, our transcriptomic data revealed that Cabo-resistant cells exhibit upregulated oxidative phosphorylation (OXPHOS) and ROS pathways while downregulating glycolysis and hypoxia responses. While most rapidly proliferating cancer cells rely on glycolysis (Warburg effect), recent reports suggest that slowly dividing tumor cells preferentially utilize OXPHOS, enabling them to better withstand therapy [86]. Resistance to anticancer drugs has been linked to high OXPHOS in multiple cancers [87], where increased ROS levels further drive cancer progression [88,89]. Notably, we also observed upregulation of the Nrf2 pathway in Cabo-resistant cells, reinforcing the role of oxidative stress resistance in therapy evasion. Our previous studies demonstrated that the c-Met signaling alleviates oxidative stress in RCC via Nrf2, promoting cancer cell survival [19]. Although our data reveal upregulation of OXPHOS and ROS pathways in resistant cells, further validation of oxidative stress markers (e.g., 8-OHdG, GSH/GSSG) in patient samples will be necessary to strengthen the clinical relevance of these findings.

In summary, our findings identify AXL as a key RTK that facilitates c-Met function and mediates therapeutic resistance against Cabo treatment in RCC. Few AXL inhibitors (like, TP-0903, Bemcentinib and others) are currently available and being tested. Targeting AXL with novel inhibitor(s) can significantly enhance the therapeutic efficacy of Cabo and may represent a promising strategy to overcome drug resistance in RCC patients. However, the universality of combining AXL inhibitors with other c-Met inhibitors and the potential toxicity challenges should be thoroughly evaluated as both the receptors are expressed in normal tissues. In addition, the therapeutic potential of AXL inhibitor(s) as a single agent can also be further evaluated and considered.

CRediT authorship contribution statement

Akash Sabarwal: Writing – review & editing, Writing – original draft, Validation, Investigation, Funding acquisition, Conceptualization. **Marc Machaalani:** Writing – review & editing. **Laxminarayan Rawat:** Investigation. **Johannes Wedel:** Investigation. **Saba Tabasum:** Investigation. **Yuzuru Sasamoto:** Investigation. **Florian Buerger:** Investigation. **Josie Ascione:** Investigation. **Marc Eid:** Investigation. **Karl Semaan:** Investigation. **Eddy Saad:** Investigation. **Yifan Yang:** Investigation. **Dongwon Lee:** Investigation. **F. Stephen Hodi:** Resources, Methodology. **Matthew L. Freedman:** Resources, Methodology. **Gwo-Shu Mary Lee:** Investigation. **Murugabaskar Balan:** Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Toni K. Choueiri:** Writing – review & editing, Funding acquisition, Conceptualization. **Soumitro Pal:** Writing – review & editing, Funding acquisition, Conceptualization.

Consent for publication

All authors have read and approved the final manuscript.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

• Dr. Pal reports research support from Exelixis. Dr. Freedman is a cofounder and shareholder of Precede Biosciences. Dr. Choueiri reports institutional and/or personal, paid and/or unpaid support for research, advisory boards, consultancy, and/or honoraria past 5 years, ongoing or not, from: Alkermes, Arcus Bio, AstraZeneca, Aravive, Aveo, Bayer, Bristol Myers-Squibb, Calithera, Circle Pharma, Deciphera Pharmaceuticals, Eisai, EMD Serono, Exelixis, GlaxoSmithKline, Gilead, HiberCell, IQVA, Infinity, Ipsen, Jansen, Kanaph, Lilly, Merck, Nikang, Neomorph, Nuscan/PrecedeBio, Novartis, Oncohost, Pfizer, Roche, Sanofi/Aventis, Scholar Rock, Surface Oncology, Takeda, Tempest, Up-To-Date, CME events (Peerview, OncLive, MJH, CCO and others), outside the submitted work.·Institutional patents filed on molecular alterations and immunotherapy response/toxicity, and ctDNA.·Equity: Tempest, Pionyr, Osel, Precede Bio, CureResponse, InnDura Therapeutics, Primium.·Committees: NCCN, GU Steering Committee, ASCO (BOD 6-2024-, ESMO, ACCRU, KidneyCan.·Medical writing and editorial assistance support may have been funded by Communications companies in part.·No speaker's bureau. Mentored several non-US citizens on research projects with potential funding (in part) from non-US sources/Foreign Components.• The institution (Dana-Farber Cancer Institute) may have received additional independent funding of drug companies or/and royaltiespotentially involved in research around the subject matter.Dr. Hodi reports grants and personal fees from Bristol-Myers Squibb, personal fees from Merck, grants and personal fees from Novartis, personal fees from Compass Therapeutics, personal fees from Apricity, personal fees from Bicara, personal fees from Checkpoint Therapeutics, personal fees from Bioentre, personal fees from Gossamer, personal fees from Iovance, personal fees from Catalym, personal fees from Immunocore, personal fees from Kairos, personal fees from Rheos, personal fees from Bayer, personal fees from Zumutor, personal fees from Corner Therapeutics, personal fees from Puretech, personal fees from Curis, personal fees from Astra Zeneca, personal fees from Pliant, personal fees from Solu Therapeutics, personal fees from Vir Biotechnology, personal fees from 92Bio, outside the submitted work; In addition, Dr. Hodi has a patent Methods for Treating MICA-Related Disorders (#20100111973) with royalties paid, a patent Tumor antigens and uses thereof (#7250291) issued, a patent Angiopoiten-2 Biomarkers Predictive of Anti-immune checkpoint response (#20170248603) pending, a patent Compositions and Methods for Identification, Assessment, Prevention, and Treatment of Melanoma using PD-L1 Isoforms (#20160340407) pending, a patent Therapeutic peptides (#20160046716) pending, a patent Therapeutic Peptides (#20140004112) pending, a patent Therapeutic Peptides (#20170022275) pending, a patent Therapeutic Peptides (#20170008962) pending, a patent THERAPEUTIC PEPTIDES Therapeutic Peptides patent number: 9402905 issued, a patent METHODS OF USING PEMBROLIZUMAB AND TREBANANIB pending, a patent Vaccine compositions and methods for restoring NKG2D pathway function against cancers Patent number: 10279021 with royalties paid, a patent Antibodies that bind to MHC class I polypeptide-related sequence A Patent number: 10106611 issued, a patent ÚNTI-GALECTIN ANTIBODY BIOMARKERS PREDICTIVE OF ANTI-IMMUNE CHECKPOINT AND ANTI-ANGIOGENESIS RESPONSES Publication number: 20170343552 pending, and a patent Antibodies against EDIL3 and methods of use thereof pending.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.canlet.2025.217984.

Data availability

The raw data generated in this study are available upon request.

References

[1] M. Li, L. Li, J. Zheng, Z. Li, S. Li, K. Wang, X. Chen, Liquid biopsy at the frontier in renal cell carcinoma: recent analysis of techniques and clinical application, Mol. Cancer 22 (2023) 37.
[2] Y. Wang, E.R. Suarez, G. Kastrunes, N.S.P. de Campos, R. Abbas, R.S. Pivetta, N. Murugan, G.M. Chalbatani, V. D'Andrea, W.A. Marasco, Evolution of cell therapy for renal cell carcinoma, Mol. Cancer 23 (2024) 8.
[3] A.D. Waldman, J.M. Fritz, M.J. Lenardo, A guide to cancer immunotherapy: from T cell basic science to clinical practice, Nat. Rev. Immunol. 20 (2020) 651–668.
[4] Z.-L. Liu, H.-H. Chen, L.-L. Zheng, L.-P. Sun, L. Shi, Angiogenic signaling pathways and anti-angiogenic therapy for cancer, Signal Transduct. Targeted Ther. 8 (2023) 198.
[5] J.H. Kim, B.J. Kim, H.S. Kim, Clinicopathological impacts of high c-Met expression in renal cell carcinoma: a meta-analysis and review, Oncotarget 8 (2017) 75478–75487.
[6] G.T. Gibney, S.A. Aziz, R.L. Camp, P. Conrad, B.E. Schwartz, C.R. Chen, W.K. Kelly, H.M. Kluger, c-Met is a prognostic marker and potential therapeutic target in clear cell renal cell carcinoma, Ann. Oncol. 24 (2013) 343–349.
[7] L.C. Harshman, T.K. Choueiri, Targeting the hepatocyte growth factor/c-Met signaling pathway in renal cell carcinoma, Cancer J. 19 (2013) 316–323.
[8] E. Gherardi, W. Birchmeier, C. Birchmeier, G.V. Woude, Targeting MET in cancer: rationale and progress, Nat. Rev. Cancer 12 (2012) 89–103.
[9] X. Huang, E. Li, H. Shen, X. Wang, T. Tang, X. Zhang, J. Xu, Z. Tang, C. Guo, X. Bai, T. Liang, Targeting the HGF/MET axis in cancer therapy: challenges in resistance and opportunities for improvement, Front. Cell Dev. Biol. 8 (2020).
[10] J.V. Abella, P. Peschard, M.A. Naujokas, T. Lin, C. Saucier, S. Urbé, M. Park, Met/hepatocyte growth factor receptor ubiquitination suppresses transformation and is required for Hrs phosphorylation, Mol. Cell Biol. 25 (2005) 9632–9645.
[11] K. Tarvestad-Laise, B.P. Ceresa, Knockout of c-Cbl/Cbl-b slows c-Met trafficking resulting in enhanced signaling in corneal epithelial cells, J. Biol. Chem. 299 (2023) 105233.
[12] J. Zhang, A. Babic, Regulation of the MET oncogene: molecular mechanisms, Carcinogenesis 37 (2016) 345–355.
[13] W.R. Critchley, C. Pellet-Many, B. Ringham-Terry, M.A. Harrison, I.C. Zachary, S. Ponnambalam, Receptor tyrosine kinase ubiquitination and De-Ubiquitination in signal transduction and receptor trafficking, Cells 7 (2018).
[14] M. Jeffers, G.A. Taylor, K.M. Weidner, S. Omura, G.F. Vande Woude, Degradation of the Met tyrosine kinase receptor by the ubiquitin-proteasome pathway, Mol. Cell Biol. 17 (1997) 799–808.
[15] T.K. Choueiri, S. Halabi, B.L. Sanford, O. Hahn, M.D. Michaelson, M.K. Walsh, D. R. Feldman, T. Olencki, J. Picus, E.J. Small, S. Dakhil, D.J. George, M.J. Morris, Cabozantinib versus sunitinib as initial targeted therapy for patients with metastatic renal cell carcinoma of poor or intermediate risk: the alliance A031203 CABOSUN trial, J. Clin. Oncol. 35 (2017) 591–597.
[16] M. Schmidinger, R. Danesi, Management of adverse events associated with cabozantinib therapy in renal cell carcinoma, Oncologist 23 (2018) 306–315.

[17] Y. Yang, S. Li, Y. Wang, Y. Zhao, Q. Li, Protein tyrosine kinase inhibitor resistance in malignant tumors: molecular mechanisms and future perspective, Signal Transduct. Targeted Ther. 7 (2022) 329.

[18] Q. Jiao, L. Bi, Y. Ren, S. Song, Q. Wang, Y.S. Wang, Advances in studies of tyrosine kinase inhibitors and their acquired resistance, Mol. Cancer 17 (2018) 36.

[19] S. Chakraborty, M. Balan, E. Flynn, D. Zurakowski, T.K. Choueiri, S. Pal, Activation of c-Met in cancer cells mediates growth-promoting signals against oxidative stress through Nrf2-HO-1, Oncogenesis 8 (2019) 7.

[20] S.L. Feldt, C.M. Bestvina, The role of MET in resistance to EGFR inhibition in NSCLC: a review of mechanisms and treatment implications, Cancers 15 (2023) 2998.

[21] N. Coleman, L. Hong, J. Zhang, J. Heymach, D. Hong, X. Le, Beyond epidermal growth factor receptor: MET amplification as a general resistance driver to targeted therapy in oncogene-driven non-small-cell lung cancer, ESMO Open 6 (2021) 100319.

[22] H. Yu, R. Liu, B. Ma, X. Li, H.Y. Yen, Y. Zhou, V. Krasnoperov, Z. Xia, X. Zhang, A. M. Bove, M. Buscarini, D. Parekh, I.S. Gill, Q. Liao, M. Tretiakova, D. Quinn, J. Zhao, P.S. Gill, Axl receptor tyrosine kinase is a potential therapeutic target in renal cell carcinoma, Br. J. Cancer 113 (2015) 616–625.

[23] A. Sabarwal, J. Wedel, K. Liu, D. Zurakowski, S. Chakraborty, E. Flynn, D. M. Briscoe, M. Balan, S. Pal, A combination therapy using an mTOR inhibitor and Honokiol effectively induces autophagy through the modulation of AXL and Rubicon in renal cancer cells and restricts renal tumor growth following organ transplantation, Carcinogenesis 43 (2021) 360–370.

[24] D.K. Graham, D. DeRyckere, K.D. Davies, H.S. Earp, The TAM family: phosphatidylserine-sensing receptor tyrosine kinases gone awry in cancer, Nat. Rev. Cancer 14 (2014) 769–785.

[25] G. Zhang, M. Wang, H. Zhao, W. Cui, Function of Axl receptor tyrosine kinase in non-small cell lung cancer, Oncol. Lett. 15 (2018) 2726–2734.

[26] O. Zajac, R. Leclere, A. Nicolas, D. Meseure, C. Marchiò, A. Vincent-Salomon, S. Roman-Roman, M. Schoumacher, T. Dubois, AXL controls directed migration of mesenchymal triple-negative breast cancer cells, Cells 9 (2020).

[27] T.M. Brand, M. Iida, A.P. Stein, K.L. Corrigan, C.M. Braverman, J.P. Coan, H. E. Pearson, H. Bahrar, T.L. Fowler, B.P. Bednarz, S. Saha, D. Yang, P.S. Gill, M. W. Lingen, V. Saloura, V.M. Villaflor, R. Salgia, R.J. Kimple, D.L. Wheeler, AXL is a logical molecular target in head and neck squamous cell carcinoma, Clin. Cancer Res. 21 (2015) 2601–2612.

[28] M. Tanaka, D.W. Siemann, Therapeutic targeting of the Gas6/Axl signaling pathway in cancer, Int. J. Mol. Sci. 22 (2021) 9953.

[29] D. Zdżalik-Bielecka, A. Poświata, K. Kozik, K. Jastrzębski, K.O. Schink, M. Brewińska-Olchowik, K. Piwocka, H. Stenmark, M. Miączyńska, The GAS6-AXL signaling pathway triggers actin remodeling that drives membrane ruffling, macropinocytosis, and cancer-cell invasion, Proc. Natl. Acad. Sci. 118 (2021) e2024596118.

[30] W. Li, X. Xiong, A. Abdalla, S. Alejo, L. Zhu, F. Lu, H. Sun, HGF-induced formation of the MET–AXL–ELMO2–DOCK180 complex promotes RAC1 activation, receptor clustering, and cancer cell migration and invasion, J. Biol. Chem. 293 (2018) 15397–15418.

[31] Y.J. Choi, J.H. Kim, J.K. Rho, J.S. Kim, C.M. Choi, W.S. Kim, J. Son, J.C. Lee, AXL and MET receptor tyrosine kinases are essential for lung cancer metastasis, Oncol. Rep. 37 (2017) 2201–2208.

[32] M. Yadav, A. Sharma, K. Patne, S. Tabasum, J. Suryavanshi, L. Rawat, M. Machaalani, M. Eid, R.P. Singh, T.K. Choueiri, S. Pal, A. Sabarwal, AXL signaling in cancer: from molecular insights to targeted therapies, Signal Transduct. Targeted Ther. 10 (2025) 37.

[33] C. Zhu, Y. Wei, X. Wei, AXL receptor tyrosine kinase as a promising anti-cancer approach: functions, molecular mechanisms and clinical applications, Mol. Cancer 18 (2019) 153.

[34] C.C. Hong, J.D. Lay, J.S. Huang, A.L. Cheng, J.L. Tang, M.T. Lin, G.M. Lai, S. E. Chuang, Receptor tyrosine kinase AXL is induced by chemotherapy drugs and overexpression of AXL confers drug resistance in acute myeloid leukemia, Cancer Lett. 268 (2008) 314–324.

[35] T.M. Brand, M. Iida, A.P. Stein, K.L. Corrigan, C.M. Braverman, N. Luthar, M. Toulany, P.S. Gill, R. Salgia, R.J. Kimple, D.L. Wheeler, AXL mediates resistance to cetuximab therapy, Cancer Res. 74 (2014) 5152–5164.

[36] Y.S. Tan, Y.L. Lei, Isolation of tumor-infiltrating lymphocytes by ficoll-paque density gradient centrifugation, Methods Mol. Biol. 1960 (2019) 93–99.

[37] M. Melter, M.E. Reinders, M. Sho, S. Pal, C. Geehan, M.D. Denton, D. Mukhopadhyay, D.M. Briscoe, Ligation of CD40 induces the expression of vascular endothelial growth factor by endothelial cells and monocytes and promotes angiogenesis in vivo, Blood 96 (2000) 3801–3808.

[38] A. Dobin, C.A. Davis, F. Schlesinger, J. Drenkow, C. Zaleski, S. Jha, P. Batut, M. Chaisson, T.R. Gingeras, STAR: ultrafast universal RNA-seq aligner, Bioinformatics 29 (2013) 15–21.

[39] H. Li, R. Durbin, Fast and accurate short read alignment with Burrows-Wheeler transform, Bioinformatics 25 (2009) 1754–1760.

[40] D.S. DeLuca, J.Z. Levin, A. Sivachenko, T. Fennell, M.D. Nazaire, C. Williams, M. Reich, W. Winckler, G. Getz, RNA-SeQC: RNA-seq metrics for quality control and process optimization, Bioinformatics 28 (2012) 1530–1532.

[41] M.I. Love, W. Huber, S. Anders, Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2, Genome Biol. 15 (2014) 550.

[42] A. Subramanian, P. Tamayo, V.K. Mootha, S. Mukherjee, B.L. Ebert, M.A. Gillette, A. Paulovich, S.L. Pomeroy, T.R. Golub, E.S. Lander, J.P. Mesirov, Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles, Proc. Natl. Acad. Sci. U. S. A. 102 (2005) 15545–15550.

[43] G. Yu, L.G. Wang, Y. Han, Q.Y. He, clusterProfiler: an R package for comparing biological themes among gene clusters, OMICS 16 (2012) 284–287.

[44] Y. Zhang, T. Liu, C.A. Meyer, J. Eeckhoute, D.S. Johnson, B.E. Bernstein, C. Nusbaum, R.M. Myers, M. Brown, W. Li, X.S. Liu, Model-based analysis of ChIP-Seq (MACS), Genome Biol. 9 (2008) R137.

[45] A.R. Quinlan, I.M. Hall, BEDTools: a flexible suite of utilities for comparing genomic features, Bioinformatics 26 (2010) 841–842.

[46] M. Lawrence, R. Gentleman, V. Carey, Rtracklayer: an R package for interfacing with genome browsers, Bioinformatics 25 (2009) 1841–1842.

[47] M. Lawrence, W. Huber, H. Pagès, P. Aboyoun, M. Carlson, R. Gentleman, M. T. Morgan, V.J. Carey, Software for computing and annotating genomic ranges, PLoS Comput. Biol. 9 (2013) e1003118.

[48] F. Ramírez, F. Dündar, S. Diehl, B.A. Grüning, T. Manke, deepTools: a flexible platform for exploring deep-sequencing data, Nucleic Acids Res. 42 (2014) W187–W191.

[49] D.S. Chandrashekar, S.K. Karthikeyan, P.K. Korla, H. Patel, A.R. Shovon, M. Athar, G.J. Netto, Z.S. Qin, S. Kumar, U. Manne, C.J. Creighton, S. Varambally, UALCAN: an update to the integrated cancer data analysis platform, Neoplasia 25 (2022) 18–27.

[50] D.S. Chandrashekar, B. Bashel, S.A.H. Balasubramanya, C.J. Creighton, I. Ponce-Rodriguez, B.V.S.K. Chakravarthi, S. Varambally, UALCAN: a portal for facilitating tumor subgroup gene expression and survival analyses, Neoplasia 19 (2017) 649–658.

[51] A.I. Minchinton, I.F. Tannock, Drug penetration in solid tumours, Nat. Rev. Cancer 6 (2006) 583–592.

[52] I.H. Bartelink, E.F. Jones, S.K. Shahidi-Latham, P.R.E. Lee, Y. Zheng, P. Vicini, L. van 't Veer, D. Wolf, A. Iagaru, D.L. Kroetz, B. Prideaux, C. Cilliers, G. M. Thurber, Z. Wimana, G. Gebhart, Tumor drug penetration measurements could be the neglected piece of the personalized cancer treatment puzzle, Clin. Pharmacol. Ther. 106 (2019) 148–163.

[53] Y. Zhao, J. Zhao, R.K. Mialki, J. Wei, E.W. Spannhake, R. Salgia, V. Natarajan, Lipopolysaccharide-induced phosphorylation of c-Met tyrosine residue 1003 regulates c-Met intracellular trafficking and lung epithelial barrier function, Am. J. Physiol. Lung Cell. Mol. Physiol. 305 (2013) L56–L63.

[54] N. Lee, J. Lee, S.H. Lee, S. Kim, S. Kim, Disproportionately high levels of HGF induce the degradation of the c-met receptor through the proteasomal degradation pathway, Biochem. Biophys. Res. Commun. 505 (2018) 925–930.

[55] J. Lefebvre, F. Ancot, C. Leroy, G. Muharram, A. Lemière, D. Tulasne, Met degradation: more than one stone to shoot a receptor down, FASEB J. 26 (2012) 1387–1399.

[56] K.C. Park, D.R. Richardson, The c-MET oncoprotein: function, mechanisms of degradation and its targeting by novel anti-cancer agents, Biochim. Biophys. Acta Gen. Subj. 1864 (2020) 129650.

[57] J. Michaelis, M. Grabbert, A. Sigle, M. Yilmaz, D. Schlager, C. Gratzke, A. Miernik, D.S. Schoeb, Tyrosine kinase inhibitors in the treatment of metastasised renal cell carcinoma-future or the past? Cancers (Basel) 14 (2022).

[58] P.C. Barata, B.I. Rini, Treatment of renal cell carcinoma: current status and future directions, CA Cancer J. Clin. 67 (2017) 507–524.

[59] X. Wang, H. Zhang, X. Chen, Drug resistance and combating drug resistance in cancer, Cancer Drug Resist 2 (2019) 141–160.

[60] T.K. Choueiri, T. Powles, M. Burotto, B. Escudier, M.T. Bourlon, B. Zurawski, V. M. Oyervides Juárez, J.J. Hsieh, U. Basso, A.Y. Shah, C. Suárez, A. Hamzaj, J. C. Goh, C. Barrios, M. Richardet, C. Porta, R. Kowalyszyn, J.P. Feregrino, J. Żołnierek, D. Pook, E.R. Kessler, Y. Tomita, R. Mizuno, J. Bedke, J. Zhang, M. A. Maurer, B. Simsek, F. Ejzykowicz, G.M. Schwab, A.B. Apolo, R.J. Motzer, Nivolumab plus cabozantinib versus sunitinib for advanced renal-cell carcinoma, N. Engl. J. Med. 384 (2021) 829–841.

[61] T.K. Choueiri, T. Powles, L. Albiges, M. Burotto, C. Szczylik, B. Zurawski, E. Yanez Ruiz, M. Maruzzo, A. Suarez Zaizar, L.E. Fein, F.A. Schutz, D.Y.C. Heng, F. Wang, F. Mataveli, Y.L. Chang, M. van Kooten Losio, C. Suarez, R.J. Motzer, Cabozantinib plus Nivolumab and ipilimumab in renal-cell carcinoma, N. Engl. J. Med. 388 (2023) 1767–1778.

[62] L. Zhou, X.D. Liu, M. Sun, X. Zhang, P. German, S. Bai, Z. Ding, N. Tannir, C. G. Wood, S.F. Matin, J.A. Karam, P. Tamboli, K. Sircar, P. Rao, E.B. Rankin, D. A. Laird, A.G. Hoang, C.L. Walker, A.J. Giaccia, E. Jonasch, Targeting MET and AXL overcomes resistance to sunitinib therapy in renal cell carcinoma, Oncogene 35 (2016) 2687–2697.

[63] M.W. Dewhirst, T.W. Secomb, Transport of drugs from blood vessels to tumour tissue, Nat. Rev. Cancer 17 (2017) 738–750.

[64] O. Trédan, C.M. Galmarini, K. Patel, I.F. Tannock, Drug resistance and the solid tumor microenvironment, JNCI: J. Natl. Cancer Inst. 99 (2007) 1441–1454.

[65] A. Wu, K. Loutherback, G. Lambert, L. Estévez-Salmerón, T.D. Tlsty, R.H. Austin, J. C. Sturm, Cell motility and drug gradients in the emergence of resistance to chemotherapy, Proc. Natl. Acad. Sci. 110 (2013) 16103–16108.

[66] Y. Laimon, M. Paul, T. Jammihal, F. Ghandour, N. El Ahmar, B. Simsek, S. Matar, T. Denize, R. Mohanna, G.R. De Oliveira, A. Delcea, S. Shukla, D. Braun, M. Sun, D. Aftab, E. Saad, M. Machaalani, W. Xie, T. Choueiri, S. Signoretti, 831 AXL Receptor Tyrosine Kinase Expression in Metastatic Renal Cell Carcinoma: Insights from the METEOR Trial, 105, Laboratory Investigation, 2025.

[67] S. Lacy, J. Nielsen, B. Yang, D. Miles, L. Nguyen, M. Hutmacher, Population exposure-response analysis of cabozantinib efficacy and safety endpoints in patients with renal cell carcinoma, Cancer Chemother. Pharmacol. 81 (2018) 1061–1070.

[68] A. Sharma, R. Elias, A. Christie, N.S. Williams, I. Pedrosa, J. Brugarolas, Potential role of dose escalation in overcoming acquired resistance to cabozantinib in metastatic renal cell carcinoma (mRCC), J. Clin. Oncol. 39 (2021) e16568.

[69] L. Rawat, M. Balan, Y. Sasamoto, A. Sabarwal, S. Pal, A novel combination therapy with Cabozantinib and Honokiol effectively inhibits c-Met-Nrf2-induced renal tumor growth through increased oxidative stress, Redox Biol. 68 (2023) 102945.
[70] T. Sen, P. Tong, L. Diao, L. Li, Y. Fan, J. Hoff, J.V. Heymach, J. Wang, L.A. Byers, Targeting AXL and mTOR pathway overcomes primary and acquired resistance to WEE1 inhibition in small-cell lung cancer, Clin. Cancer Res. 23 (2017) 6239–6253.
[71] J. Sarantopoulos, G. Fotopoulos, F.Y.C. Tsai, M.S. Beg, A.A. Adjei, Y. Lou, M. Seetharam, M.A. Villalona-Calero, J. Melear, M. Janat-Amsbury, H. Beever, L. Mouritsen, M. Wade, B.V. Bryan, D.J. Bearss, 460P - a phase Ia/b first-in-human, open-label, dose-escalation, safety, PK and PD study of TP-0903 in solid tumours, Ann. Oncol. 30 (2019) v172.
[72] L. Feng, K. Zhao, L. Sun, X. Yin, J. Zhang, C. Liu, B. Li, SLC7A11 regulated by NRF2 modulates esophageal squamous cell carcinoma radiosensitivity by inhibiting ferroptosis, J. Transl. Med. 19 (2021) 367.
[73] H. Zhang, T. Ma, X. Wen, J. Jiang, J. Chen, J. Jiang, J. Xie, T. Mo, R. Li, H. Xie, G. Liang, L. Wang, Z. Zheng, X. Huang, C. Liu, Y. Baihetiyaer, A. Abulimiti, X. He, Z. Chen, T. Hu, W. Pan, SIK1 promotes ferroptosis resistance in pancreatic cancer via HDAC5-STAT6-SLC7A11 axis, Cancer Lett. 623 (2025) 217726.
[74] M. Balan, E. Mier y Teran, A.M. Waaga-Gasser, M. Gasser, T.K. Choueiri, G. Freeman, S. Pal, Novel roles of c-Met in the survival of renal cancer cells through the regulation of HO-1 and PD-L1 expression, J. Biol. Chem. 290 (2015) 8110–8120.
[75] H. Li, Z. Liu, L. Liu, H. Zhang, C. Han, L. Girard, H. Park, A. Zhang, C. Dong, J. Ye, A. Rayford, M. Peyton, X. Li, K. Avila, X. Cao, S. Hu, M.M. Alam, E.A. Akbay, L. M. Solis, C. Behrens, S. Hernandez-Ruiz, W. Lu, I. Wistuba, J.V. Heymach, M. Chisamore, D. Micklem, H. Gabra, G. Gausdal, J.B. Lorens, B. Li, Y.X. Fu, J. D. Minna, R.A. Brekken, AXL targeting restores PD-1 blockade sensitivity of STK11/LKB1 mutant NSCLC through expansion of TCF1(+) CD8 T cells, Cell Rep. Med. 3 (2022) 100554.
[76] S. Gusenbauer, P. Vlaicu, A. Ullrich, HGF induces novel EGFR functions involved in resistance formation to tyrosine kinase inhibitors, Oncogene 32 (2013) 3846–3856.
[77] S.M. Dieter, D. Lovecchio, A. Pataskar, M.K. Zowada, P.R. Körner, A. Khalizieva, O. van Tellingen, D. Jäger, H. Glimm, R. Agami, Suppression of heparan sulfation re-sensitizes YAP1-driven melanoma to MAPK pathway inhibitors, Oncogene 41 (2022) 3953–3968.
[78] E.B. Rankin, K.C. Fuh, L. Castellini, K. Viswanathan, E.C. Finger, A.N. Diep, E. L. LaGory, M.S. Kariolis, A. Chan, D. Lindgren, H. Axelson, Y.R. Miao, A.J. Krieg, A. J. Giaccia, Direct regulation of GAS6/AXL signaling by HIF promotes renal metastasis through SRC and MET, Proc. Natl. Acad. Sci. 111 (2014) 13373–13378.
[79] M.Z. Xu, S.W. Chan, A.M. Liu, K.F. Wong, S.T. Fan, J. Chen, R.T. Poon, L. Zender, S. W. Lowe, W. Hong, J.M. Luk, AXL receptor kinase is a mediator of YAP-dependent oncogenic functions in hepatocellular carcinoma, Oncogene 30 (2011) 1229–1240.
[80] Z. Zhang, J.C. Lee, L. Lin, V. Olivas, V. Au, T. LaFramboise, M. Abdel-Rahman, X. Wang, A.D. Levine, J.K. Rho, Y.J. Choi, C.M. Choi, S.W. Kim, S.J. Jang, Y. S. Park, W.S. Kim, D.H. Lee, J.S. Lee, V.A. Miller, M. Arcila, M. Ladanyi, P. Moonsamy, C. Sawyers, T.J. Boggon, P.C. Ma, C. Costa, M. Taron, R. Rosell, B. Halmos, T.G. Bivona, Activation of the AXL kinase causes resistance to EGFR-targeted therapy in lung cancer, Nat. Genet. 44 (2012) 852–860.
[81] N. Karachaliou, I. Chaib, A.F. Cardona, J. Berenguer, J.W.P. Bracht, J. Yang, X. Cai, Z. Wang, C. Hu, A. Drozdowskyj, C.C. Servat, J.C. Servat, M. Ito, I. Attili, E. Aldeguer, A.G. Capitan, J. Rodriguez, L. Rojas, S. Viteri, M.A. Molina-Vila, S. I. Ou, M. Okada, T.S. Mok, T.G. Bivona, M. Ono, J. Cui, Y.C.S. Ramón, A. Frias, P. Cao, R. Rosell, Common Co-activation of AXL and CDCP1 in EGFR-mutation-positive non-smallcell lung cancer associated with poor prognosis, EBioMedicine 29 (2018) 112–127.
[82] S.L. Organ, M.S. Tsao, An overview of the c-MET signaling pathway, Ther. Adv. Med. Oncol. 3 (2011) S7–s19.
[83] M.A. Miller, M.J. Oudin, R.J. Sullivan, S.J. Wang, A.S. Meyer, H. Im, D. T. Frederick, J. Tadros, L.G. Griffith, H. Lee, R. Weissleder, K.T. Flaherty, F. B. Gertler, D.A. Lauffenburger, Reduced proteolytic shedding of receptor tyrosine kinases is a post-translational mechanism of kinase inhibitor resistance, Cancer Discov. 6 (2016) 382–399.
[84] R.M. Linger, A.K. Keating, H.S. Earp, D.K. Graham, TAM receptor tyrosine kinases: biologic functions, signaling, and potential therapeutic targeting in human cancer, Adv. Cancer Res. 100 (2008) 35–83.
[85] L. Brandão, J. Migdall-Wilson, K. Eisenman, D.K. Graham, TAM receptors in leukemia: expression, signaling, and therapeutic implications, Crit. Rev. Oncog. 16 (2011) 47–63.
[86] Z. Zhao, Y. Mei, Z. Wang, W. He, The effect of oxidative phosphorylation on cancer drug resistance, Cancers (Basel) (2022) 15.
[87] C. Uslu, E. Kapan, A. Lyakhovich, Cancer resistance and metastasis are maintained through oxidative phosphorylation, Cancer Lett. 587 (2024) 216705.
[88] K. Tuy, L. Rickenbacker, A.B. Hjelmeland, Reactive oxygen species produced by altered tumor metabolism impacts cancer stem cell maintenance, Redox Biol. 44 (2021) 101953.
[89] A.L. Furfaro, N. Traverso, C. Domenicotti, S. Piras, L. Moretta, U.M. Marinari, M. A. Pronzato, M. Nitti, The Nrf2/HO-1 axis in cancer cell growth and chemoresistance, Oxid. Med. Cell. Longev. 2016 (2016) 1958174.